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Chapter Contents

III. PREFACE

The FutuRaM project aims to quantify the current and future availability of secondary raw materials (SRM), focusing on critical raw materials (CRMs) [80]. This study is concerned with six waste streams in the EU member states, as well as Iceland, Norway, Switzerland, and the United Kingdom (EU27+4). In this report, the EU27+4 will henceforth be referred to as the EU, unless specified otherwise.

THE WASTE STREAMS COVERED IN FUTURAM ARE:

- **Waste batteries (BAT)**
- **Construction and demolition waste (CDW)**
- **End-of-life vehicles (ELV)**
- **Mining waste (MIN)**
- **Slags and ashes (SLASH)**
- **Waste electrical and electronic equipment (WEEE)**

Work package two (WP2) is conducting foresight studies for materials that are either classified as critical to the EU economy or are significant due to factors such as their large volumes, commercial importance, and environmental impacts [80, 1, 22, 141]. WP2 is tasked with developing a set of coherent scenarios for material use and waste/recovery over time across various sectors in the EU. This report describes the three distinct scenarios and the process by which they were developed.

THE THREE SCENARIOS THAT HAVE BEEN DEVELOPED IN FUTURAM ARE:

- BAU **I. Business-as-usual (BAU)**
- **II. Recovery (REC)**
- **III. Circularity (CIR)**

IV. REPORT CONTENTS

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VII. EXECUTIVE SUMMARY

This report presents the first phase of the scenario development process — the storyline narrative phase. Three distinct future scenarios have been drafted up to the year 2050: Business as Usual, Recovery, and Circularity. These scenarios are designed to be internally consistent and provide an overview of the potential future landscape of waste management and SRM recovery within the EU.

The scenario development process employs a methodology that integrates both forecasting and back-casting techniques to build a comprehensive, future-facing knowledge base that can aid fact-based decision-making [81, 23, 82, 83, 76, 84, 85].

In the next phases of scenario development, future product composition and recovery technology will be assessed, scenario elements will be quantified, and all data will be integrated with the quantitative models for waste generation and SRM recovery.

The FutuRaM project aims to offer a nuanced understanding of the potential future waste management and resource recovery landscape within the EU. This approach provides insights into key drivers, uncertainties, and the possible impacts of policy and technological advancements. Additionally, by aligning SRM recovery efforts with the United Nations Framework Classification for Resources (UNFC) [150], the project aims to facilitate the commercial exploitation of SRMs and CRMs by manufacturers, recyclers, and investors. With the comprehensive knowledge base that we are developing, FutuRaM aims to support informed decision-making by policymakers and government, as well as industry and community stakeholders.

FUTURAM'S THREE FUTURE SCENARIOS

Scenario I: Business-as-usual (BAU)

The BAU scenario extrapolates current trends into the future with limited change. Using forecasting techniques, it projects a potential future where there are minor advancements in resource efficiency, recovery technology, and the energy transition, but primary extraction of raw materials remains the dominant practice.

Scenario II: Recovery (REC)

The Recovery scenario imagines a future leveraging advanced technology to significantly enhance SRM recovery from waste streams. It outlines a future where the EU successfully meets its recycling and recovery targets through an effective waste management system and circular design principles [24, 86]. This scenario sees an increased recovery rate of SRMs, extensive use of digitalisation and automation in recycling processes, and new or strengthened waste regulations in line with EU targets.

Scenario III: Circularity (CIR)

The Circularity scenario captures the ideal of a fully realised circular economy, going beyond end-of-life recovery to minimise waste at every production and consumption stage. It predicts a future where the EU's targets for recycling, recovery, and circularity are met through extensive stakeholder collaboration, new business models, and increased use of renewable energy and circular economy technologies [2, 87, 88].

VIII. OVERVIEW OF THE SCENARIO STORYLINES

Scenario I: Business-as-usual (BAU)

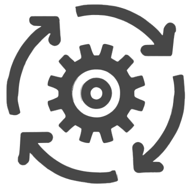
See section 3.1 for the full scenario description and waste-stream-specific scenario impact narratives.

This scenario envisions the future based on the current situation, extending to 2050 with very little deviation from present consumption patterns and without substantial development of the secondary raw material (SRM) recovery system. While there may be advances in some areas such as resource efficiency, recovery technology, and the energy transition, substantial modifications remain hindered by economic, social, and political constraints. The extraction of primary raw materials continues to be the predominant source utilised to satisfy the EU's growing SRM demand.

In the Business as usual (linear economy) scenario, the following are key characteristics:

- A forecasting model is used to predict the future based on the current situation and the development of existing trends.
- EU targets including those for eco-design, recycling and recovery are not met, and the current linear model largely persists.
- Material demand remains coupled with economic growth, perpetuating a trend of increasing consumption.
- Primary mining and extraction persist as the leading sources of raw materials, underlining the dependency on traditional extraction methods.
- Recycling and recovery rates continue to lag, leading to increased production of SRM-containing waste that signals missed opportunities for resource reuse.
- The EU's dependency on imports of SRMs escalates, heightening the risk of supply disruptions [25].
- Investment in new SRM recovery technologies remains minimal, stifling innovation and advancements in this field.
- The industrial focus remains on cost-effective material production and use, disregarding the long-term sustainability aspect.
- Material scarcity and price fluctuations pose potential risks to the EU industry, highlighting the vulnerability of this business model [26].
- Without any significant updates to environmental regulations, the negative impacts on ecosystems and biodiversity intensify.
- Mining activity in the EU remains limited and concentrated in only a few member states. Current exploration projects (e.g., for Lithium in PT, FR, UK and rare earths in SE) are not realised.
- The transitions to renewable energy and e-mobility continue at their current pace.

Scenario II: Recovery (BAU)



See section 3.2 for the full scenario description and the waste-stream-specific scenario impact narratives.

In the recovery scenario, the central emphasis is on harnessing sophisticated technologies to salvage SRMs from waste streams at the end of their lifecycle. While there are noticeable strides towards the incorporation of 'circular design' principles and re-X strategies (which focus on reducing, reusing, recycling, repairing, and refurbishing), material demand increases similarly to the BAU scenario. This is, however, mitigated to some extent by the implementation of a comprehensive material recovery system.

Key features of this technology-promoted recovery scenario include:

- This scenario uses a combination of forecasting and backcasting methods to envision the future.
- The backcasting method is used for scenario factors that are covered by governmental targets, starting with the desired outcome and working backwards to the present.
- The forecasting method is used for scenario factors that are not covered by governmental targets, starting with the current situation and extending to the future.
- EU targets for recycling and recovery are met, due to the EU's waste management system becoming more expansive, efficient and effective.
- Technological innovation drives increased recovery rates of SRMs, enabling the more efficient use of waste.
- Digitalisation and automation are more extensively used in recycling processes, leading to enhanced productivity and efficiency.
- Business models like leasing and take-back schemes emerge, altering traditional consumption patterns (here, the focus is on take-back for recycling).
- Ecodesign mandates are implemented, again, here, with a focus on end-of-life recovery.
- There is greater exploration and exploitation of alternative sources such as urban mining, waste streams, and tailings, presenting novel opportunities for resource acquisition.
- New waste regulations and guidelines for SRM recovery are implemented, enforcing better management and extraction of SRMs.
- Investment in research and development for SRM recovery technologies experiences an upswing, promoting continuous innovation in this field.
- Closer collaboration and information sharing between industry and government institutions (e.g., waste tracking and digital product passports) streamline processes and expedite decision-making.
- New jobs are created in the recycling and recovery sector, offering economic benefits and improving overall employment rates.
- SRM production and use become more efficient and cost-effective, fostering economic sustainability.

Scenario III: Circularity

See section 3.3 for the full scenario description and the waste-stream-specific scenario impact narratives.

In this scenario, we move in the direction of the maximum achievable state of material efficiency as government policy, private innovation and social changes are rapidly driving the transition toward a circular economy. The emphasis here rests heavily on re-X strategies that are implemented in the design phase of products (e.g., repairability and re-manufacturability) and that are actualised by changes in consumer behaviour (e.g reduction, refusal, engagement in the 'sharing economy' and curtailment of the 'throw-away' mindset).

Further, being enabled by the widespread adoption of 'circular design' principles and improvements in information transparency (e.g., waste tracking and digital product passports) the system for the treatment of post-consumer waste can divert a significant amount of their inflows (to, for example, re-use and re-manufacture) with the residual fraction being readily segregated into purer, more efficiently recoverable, material streams.

This scenario envisions a future where government policies are in synergy with private sector innovation and societal changes, driving a wholesale transition towards a circular economy. Unlike the recovery scenario, where the focus is on the end-of-life recovery of materials, this scenario emphasises minimising waste at all stages, starting from the design phase itself.

The circular economy scenario is characterised by the following:

- This scenario uses a combination of forecasting and backcasting methods to envision the future.
- The backcasting method is used for scenario factors that are covered by governmental targets, starting with the desired outcome and working backwards to the present.
- The forecasting method is used for scenario factors that are not covered by governmental targets, starting with the current situation and extending to the future.
- EU targets for recycling and recovery are met, as are those for circularity, due to advances in waste management, ecodesign and re-X strategies.
- A circular economy is implemented, prioritising waste reduction, resource efficiency, and a shift from the 'take-make-dispose' model.
- A notable increase in SRM recycling and recovery rates, indicating an efficient use of resources.
- A larger emphasis on designing products for reuse and recycling, making waste a valuable resource rather than a problem.
- More extensive use of renewable energy and clean technologies in SRM production and use, supporting a low-carbon economy.
- Collaboration between stakeholders — including industry, government, and consumers — improves, enhancing the implementation of circular practices.
- New business models like leasing and take-back schemes emerge, altering traditional consumption patterns [89].
- Digitalisation and data use are heightened to improve efficiency and traceability, aiding in effective resource management.

- Investment in research and development for circular economy technologies increases, driving innovation and adoption.
- Awareness and education around sustainable consumption and production practices are amplified, leading to behavioural changes in society.
- Reliance on imports decreases, suggesting greater self-sufficiency and sustainability.
- The creation of new jobs within the recycling, recovery and re-X sectors boosts the economy and alleviates social inequality.
- Stricter waste regulations and product design guidelines are introduced, accelerating the transition towards circularity.

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IX. ABBREVIATIONS

Table 1.2: List of abbreviations

ACRONYM	DEFINITION
AI	Artificial Intelligence
BAU	Business as Usual
BATT	Waste Batteries
BRGM	French Geological Survey (Bureau de Recherches Géologiques et Minières)
CDW	Construction and Demolition Waste
CE	Circular Economy
CRM	Critical Raw Material
CU	Chalmers University
EEE	Electrical and Electronic Equipment
ELV	End-of-Life Vehicles
Empa	Swiss Federal Laboratories for Materials Science and Technology
EoL	End-of-Life
EoU	End-of-Use
EoW	End-of-Waste
EU	European Union
EU27+4	EU + Iceland, Norway, Switzerland and the United Kingdom
EPR	Extended Producer Responsibility
GDP	Gross Domestic Product
GEC	Global Energy and Climate [Model]
GTK	Geological Survey of Finland
LCA	Life Cycle Assessment
LCC	Life Cycle Cost Assessment
LMU	Ludwig Maximilian University of Munich
LU	Leiden University
MIN	Mining Waste
R&D	Research and Development
REACH	Registration, Evaluation, Authorization, and Restriction of Chemicals
RECHARGE	EU rechargeable battery industry association
SGU	Geological Survey of Sweden
SLASH	Slags and Ashes
S-LCA	Social Life Cycle Assessment
SLCA	Sustainability Life Cycle Assessment
SRM	Secondary Raw Material
TRL	Technology Readiness Level
TUB	Technische Universität Berlin
UCL	University College London
UNITAR	United Nations Institute for Training and Research

Continued on next page

Table 1.2 — Continued from previous page

ACRONYM	DEFINITION
UNFC	United Nations Framework Classification for Resources
VITO	Flemish Institute for Technological Research
WEEE	Waste Electrical and Electronic Equipment
WEEE Forum	Waste Electrical and Electronic Equipment Forum
WFD	Waste Framework Directive

X. TERMINOLOGY (ABBREVIATED)

The following table provides an abbreviated list of terminology used in this report. See section 13.1 for a complete list.

Table 1.3: List of terminology (abbreviated)

TERM	DEFINITION
Backcasting	A method for predicting future trends based on a desired future state.
Business-as-usual	A scenario that assumes no significant changes in current trends and policies.
Circular economy	An economic system that prioritises waste reduction and resource efficiency.
Critical Raw Material	A raw material that is economically and strategically important to the EU, but with a high risk of supply disruption.
Forecasting	A method for predicting future trends based on historical data.
Recovery	The process of recovering SRMs from waste streams.
Re-X	A general term for circular strategies such as reuse, repair, refurbishment, remanufacturing and recycling.
Scenario	A plausible and coherent description of how the future may develop based on a set of assumptions.
Secondary Raw Material	A material that has been recovered from waste and can be used as a substitute for a primary raw material.
Storyline	A qualitative description of a scenario, including the key drivers, actors and events.

XI. DESCRIPTION OF FUTURAM WORK PACKAGE 2

A full breakdown of the tasks and subtasks in WP2, along with the responsible partners is provided in section 13.9. The following sections provide a brief description of each task.

More information can be found in the grant agreement [151], the consortium agreement [152], the project management plan [153] and the Milestone 6 report [154].

DELIVERABLE

D2.1: Report on the environmental and socioeconomic barriers to SRM recovery — *Month 47: May 2026*

XI.1. Objectives

WP2 will conduct foresight studies for materials critical to the EU economy, or materials that have significant impacts on EU sustainability because of their large volumes. WP2 will develop a set of coherent scenarios for material use and waste/recovery over time in various sectors in the EU: WEEE, ELV, BAT, CDW, MINW, SLASH.

Context

Source: [151]

MODELLING THE FUTURE OF WASTE GENERATION AND SRM RECOVERY

In WP2, modelling the foresight requires dealing with much unknown information and developments. A convincing mathematical model on the future thus requires a strong narrative developed from stakeholders and existing literature regarding how future circular behaviours, recycling and recovery technologies, and the overall material economy will develop. Furthermore, if the mathematical model used is too detailed, there will be many data gaps, leading to it being impractical to use and potentially leading to unrealistic results. This means a good balance needs to be found between data availability and its translation into a quantification of future narratives. The narratives applied to each scenario will follow plausible developments by taking into account stated MS policies by each regarding the material economy (with a special emphasis on the waste and recycling stages) and optimistic outlooks of both recycling technology using learning curves, and of increasing circular behaviour following global best practice. The rate of development towards each of these scenarios will be used for sensitivity and uncertainty analyses, such that a measure of the variability within each scenario is established.

OPEN SCIENCE

Considering the multidisciplinary character of FutuRaM and its aim to provide consistent and robust data, procedures, models, and methods, a critical discussion, harmonisation and integration of the concepts, perspectives and is crucial for the success of the project.

The consortium is committed to making the data available in open formats during the project and free of charge for the EC and all stakeholders to use and publish, along with other relevant reports tailored for the use of the EC and respecting FutuRaM's open science principles.

GENDER

Within the FutuRaM, project no specific population group will be targeted. In contrast, the consortium is aware that research often has a diversity problem since many groups are underrepresented, e.g. women, ethnic minorities, people with disabilities and socially disadvantaged populations and we will consider specific measures that will help to address specifically these groups.

We will especially consider the involvement of a variety of stakeholders in WP7.

In work packages 2, 3 and 5 we will use Delphi panels, which have an equal representation of gender and an appropriate age distribution that encapsulates multiple perspectives. In the modelling of WP2 and 4 (foresight and stock and flow models), consumption of household electronics may increase with increasing gender equality, and behavioural aspects of waste separation which could be an aspect of foresight of stock and flows.

XII. TASK DESCRIPTIONS

Task 2.1. **Develop scenario storyline (ULEI, TUB, Empa, Chalmers, WEEE Forum, BRGM, UNITAR, SGU) (M01-M18):** This task involves scanning, mapping, and assessing scenarios used in the grey, scientific, policy, and industry literature/reporting for the different waste streams, (e.g., the Shared Socioeconomic Pathways, the International Resource Panel Scenarios, the International Energy Agency Scenarios, etc) to develop cogent storylines for the three planned scenarios. These will cut across sectors and will be used for the Stock-Flow models (WP4) and will include the translation of general concepts such as stated policies, sustainable development, circular economy, to each sector. FutuRaM will develop at minimum three scenarios (1. Sustainability, 2. Recoverability, and 3. Business-as-usual).

Task 2.2. **Integrate future technologies into the scenarios (Chalmers, ULEI, TUB, Empa, WEEE Forum, BRGM, UNITAR, UCL, LMU, SGU, VITO) (M03-M20):** This task will review current and emerging technologies used in the various sectors for product manufacturing and end-of-life handling, with a special emphasis on material production, use, and recycling. Together with the storylines developed in Task 2.1, it will adapt the market share of these technologies for each sector to determine the future development of each sector.

Task 2.3. **Forecast material composition and products for each scenario (TUB, ULEI, UNITAR, Chalmers, BRGM, Empa, VITO) (M7-M20):** Following the scenarios from T2.1, the material compositions and future products for each sector will be determined based on the product and commodity demand and technology realisation (T2.2). This task will be coupled to the data collection in WP3 and WP4.

Task 2.4. **Quantify environmental and socioeconomic impacts of SRM recovery under each scenario (ULEI, TUB, Empa, UNITAR, WEEE Forum, BRGM, UCL, LMU) (M18-M36):** This task will use the information generated in Tasks 2.1-2.3, together with the material flow analysis from WP4, to quantify the future environmental and socioeconomic feedbacks for each waste sector and scenario according to future recovery technology.

Task 2.5. **Assess the environmental and socioeconomic impacts and bottlenecks of future SRM recovery (ULEI, TUB, Empa, UNITAR, Chalmers, UNITAR. WEEECycling) (M37-M47):** This task will develop a report based on an assessment on the pressures and bottlenecks associated with environmental and socioeconomic issues related to each waste sector, including the associated changes and impacts on imports and of primary raw materials production (D2.1).

Methodology

CHAPTER CONTENTS

2.1. THE CONCEPTUAL FRAMEWORK FOR SCENARIO DEVELOPMENT

The conceptual framework for scenario development is based on the following principles.

The scenarios should:

- Be based on the best available scientific knowledge and data.
- Provide a coherent and consistent picture of possible futures.
- Provide decision makers with knowledge related to the possible consequences of their decisions.
- Consider a range of plausible future outcomes, accounting for uncertainties and alternative trajectories.
- Be developed in a participatory and collaborative manner, involving relevant stakeholders and experts.
- Be transparent and well-documented, allowing for replication and further analysis (e.g., publication in peer-reviewed journals and open-access repositories).
- Be flexible and adaptable, allowing for updates and adjustments as new information becomes available.
- Consider the interconnections and interactions between different sectors, waste streams, and policy domains.
- Take into account the broader societal, economic, and environmental context in which the waste streams operate.
- Incorporate a long-term perspective, considering the potential impacts and implications over several decades.
- Capture both quantitative and qualitative aspects, integrating data-driven modelling with qualitative narratives and storylines.
- Be regularly reviewed and updated to reflect evolving knowledge, technological advancements, and policy developments.
- Be used as a tool for learning and exploration, encouraging dialogue and collaboration among stakeholders.
- Inform policy and decision-making processes, providing insights into the potential consequences of different choices and interventions.
- Be communicated effectively to a wide range of audiences, ensuring accessibility and clarity of information.
- Contribute to the advancement of knowledge and understanding in the field of waste management, resource recovery, and circular economy.

By adhering to these principles, the FutuRaM project aims to develop robust, informative, and policy-relevant scenarios that support sustainable decision-making and contribute to the transition towards a more circular and resource-efficient economy.

2.2. SCENARIO STORYLINE DEVELOPMENT PROCESS

Building scenarios involves several steps and various methodologies, these will differ depending on the specific context and objectives of the exercise [81, 23, 82, 83, 76, 90, 91, 92, 93].

The following section provides an overview of the scenario development process used in FutuRaM. Figure 2.1 provides a visual representation of the process.

2.2.1. Step 1: Define the scope and objectives

Scope and objectives of the scenario development process

The scope and objectives of the scenario development process are defined in the context of the overall aim, scope, and objectives of the FutuRaM project.

Aim of FutuRaM:

FutuRaM will develop the Secondary Raw Materials knowledge base on the availability and recoverability of secondary raw materials (SRMs) within the European Union (EU), with a special focus on critical raw materials (CRMs). The project research will enable fact-based decision-making for the recovery and use of SRMs within and outside the EU, and disseminate the data generated via an accessible knowledge base developed in the project [152, 153].

Scope of FutuRaM:

FutuRaM will establish a methodology, reporting structure, and guidance to improve the raw materials knowledge base up to 2050. FutuRaM will focus on six waste streams: batteries; electrical and electronic equipment; vehicles; mining; slags and ashes; and construction and demolition.

It will integrate SRM and CRM data to model their current stocks and flows and consider economic, technological, geopolitical, regulatory, social and environmental factors to further develop, demonstrate, and align SRM recovery projects with the United Nations Framework Classification for Resources (UNFC) [150].

This will enable the commercial exploitation of SRMs and CRMs by manufacturers, recyclers, and investors, and the knowledge base developed in the project will support policymakers and governmental authorities.

Selected objectives of the FutuRaM project are presented in Table 2.1 [152, 153].

Scope of Work Package 2 (WP2):

Given this context, the scope of the scenario development process is to develop a set of plausible scenarios that explore the future of waste management, resource recovery, and circular economy in the EU.

The scenarios will be used to identify key drivers and uncertainties that will influence the future of waste management and resource recovery. The scenarios will also be used to evaluate the potential impacts of different policy interventions and technological advancements.

Thematic scope

The scenarios will be centred on the six waste streams of FutuRaM: WEEE, ELV, BAT, CDW, MIN, and SLASH. Additionally, consideration will be given to sectors and policy domains that are relevant to these waste streams and the general context of the system. These include manufacturing, energy, and transportation, as well as policies related to the environment, the economy, society, technology, and geopolitics.

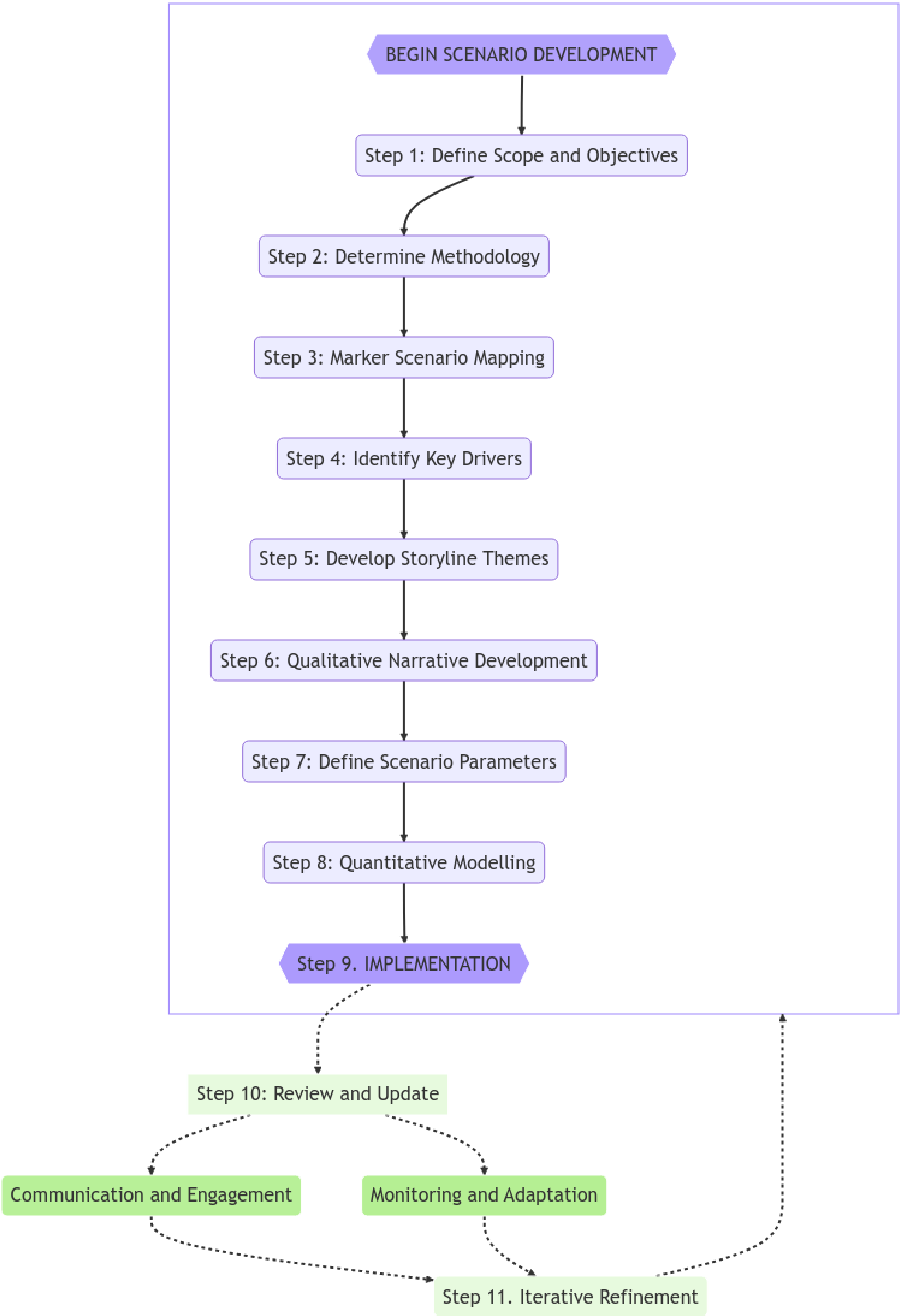

Figure 2.1: Scenario storyline development process

Table 2.1: Selected objectives of the FutuRaM project

NEED	ACTION
A successful transition to a climate-neutral, circular and digitised EU economy relies heavily on a secure supply of raw materials.	FutuRaM will quantify the future availability of SRMs for three future scenarios for the EU material economy. Forecast material demand, SRM supply for each scenario, and raw material imports to evaluate EU material autonomy.
Presently, several socioeconomic scenarios have been developed at national, EU, and/or global scales to assess the energy and mobility transition. Still lacking are specific scenarios for the SRM and CRM recovery systems	FutuRaM will develop stock-flow models for six waste streams based on holistic scenarios to map current and future material use in the economy of the EU-27 plus Iceland, Norway, Switzerland and the United Kingdom. FutuRaM will extend existing model approaches by a set of distinct scenarios that cover circular economy, high SRMs recoverability, and business as usual.

Geographic scope

The scenarios will be developed for the EU-27 plus Iceland, Norway, Switzerland and the United Kingdom (EU27+4). The scenarios will consider the current and future waste management practices and resource recovery technologies in these countries. Additionally, the scenarios will consider the current and future policies and targets related to waste management and resource efficiency in these countries. To some extent, the scenarios will also consider the current and future trade relationships between these countries and other countries around the world.

Temporal scope

The scenarios will be developed for the time horizon of 2025–2050. This time horizon is aligned with the long-term targets of the EU, including the EU Green Deal, the EU Circular Economy Action Plan, and the EU Industrial Strategy.

The discrete stages in the forecasts are planned to be: 2025, 2030, 2035, 2040, 2045 and 2050.

The temporal resolution of the scenarios will be determined during the quantification phase of the scenario development process.

While it is possible to develop scenarios with a high (or even continuous) temporal resolution, that of these scenarios will be determined based on the availability and quality of data. It is important to acknowledge that providing too high a temporal resolution may lead to a false sense of accuracy and precision.

Furthermore, the scenarios will be developed with the understanding that the further into the future we look, the more uncertain the predictions become [83, 94, 95].

Consideration of EU legislation and policy targets

The scenarios developed in FutuRaM consider targets that the EU is setting for specific elements, materials, and waste streams. The targets incorporated into FuTuRaM's scenarios are aligned with the ambitions of the EU's Green Deal [3] and its Critical Raw Materials (CRM) act [1].

Additionally, the consumer-product-centric waste streams BATT, ELV, and WEEE have specific EU legislation that is directly applicable to them and will be considered in detail in the scenarios [4, 5, 6, 142, 7,

WP2 — Aims and objectives definition

Table 2.2: FutuRaM WP2 aims and objectives

AIM	OBJECTIVE
Quantifying the current and future availability of secondary raw materials (SRM), particularly critical raw materials (CRM), for the identified waste streams from 2025 until 2050.	Developing a set of plausible scenarios that encompass these waste streams and provide quantitative estimates of the current and future availability of SRM and CRMs.
Informing private and public sector decision-making processes by assessing the impacts of different legislative and policy strategies related to waste management and resource efficiency	The scenarios will cover a range of such strategies, grouped in coherent sets in each of the three storylines including recycling, reuse, remanufacturing, and landfilling. Integration of the scenario with the system model will allow assessment of the impacts of these strategies not only on the availability of SRM and CRMs, but also on the environment, the economy, and society.

8].

General policies and legislation

The EU Green Deal [3] is a set of policy initiatives by the European Commission with the overarching aim of making the EU climate-neutral in 2050.

This policy portfolio is a response to the Paris Agreement and the United Nations Sustainable Development Goals. It covers a wide range of economic sectors with an emphasis on investments towards building local, 'sustainable' industries.

The scope of FutuRaM is aligned with the EU Green Deal's goal of ensuring the sustainable sourcing and use of raw materials, reducing dependency on imports, and promoting resource security. These goals can conflict with each other; however, the modelling in FutuRaM will explore the trade-offs between them (e.g., optimising local sourcing may result in higher negative externalities).

The EU Circular Economy Action Plan [2] is a policy framework developed by the European Commission to promote the circular economy in the European Union.

It sets out a comprehensive set of measures and targets to improve resource efficiency, reduce waste, and foster sustainable production and consumption. The Action Plan includes initiatives related to product design, waste management, recycling, and resource efficiency, among others. The Action Plan is a key element of the European Green Deal and is closely linked to the EU Industrial Strategy.

The Circular Economy Action Plan:

- Aims to promote the transition to a more circular economy in the EU.
- Sets out a range of measures to promote the sustainable use of resources, reduce waste, and increase recycling.
- Includes proposals for new legislation, such as an EU-wide framework for the circular economy, and revisions to existing legislation, such as the WEEE Directive.
- Emphasises the importance of product design for the circular economy and proposes measures to promote eco-design and repairability.
- Includes initiatives to promote the use of secondary raw materials, such as the establishment of a European Raw Materials Alliance.
- Aims to reduce greenhouse gas emissions and improve resource efficiency in the EU.
- Calls for increased cooperation and dialogue among stakeholders in the circular economy.

The Critical Raw Materials Act (CRM Act) [1] is an EU regulation that aims to ensure a secure and sustainable supply of raw materials to the EU.

The Act identifies a list of strategic raw materials, which are crucial to technologies important to Europe's green and digital ambitions and for defence and space applications, that are subject to potential supply risks. The regulation will cover the entire raw materials value chain, from primary extraction to manufacturing to potential recovery as a secondary raw material.

For example: According to the CRM act, by 2030, a single 'third country' (ex-EU, ex-Schengen) should produce no more than 65% of the EU's annual consumption of each strategic raw material.

Clear benchmarks have been set for the domestic capacities of the EU in 2030:

- Extract at least 10% of the EU's annual consumption
- Process at least 40% of the EU's annual consumption
- Recycle at least 15% of the EU's annual consumption

These benchmarks have been included in the scenarios developed in FutuRaM. Specifically, in the Recovery scenario (REC), where the emphasis is on the recovery of materials from waste streams, and the Circularity scenario (CIR) where the emphasis is on the implementation of 're-X' strategies, such as recycling, remanufacturing, and reuse.

Many of these targets, benchmarks and mandates — despite being included in legislation — are considered too optimistic to be included in the Business-as-usual scenario (BAU) as they often make expectations whose attainment is likely highly unrealistic without radical reform of the waste management system, For example, the targets in the Battery Act suggest near-complete recovery for several elements [8].

Extent of policy and legislation inclusion in the scenarios

The targets that result from the planned and ongoing review processes are non-negotiable and legally binding and thus should be incorporated into our scenarios. These targets, however, are only applicable to post-consumer products, namely WEEE, BAT and ELV. This envisioned future in which legally binding targets for collection, reuse and/or material recycling are achieved can be implemented as the Recovery scenario.

If there are no targets set for a specific consumer product category, then approach targets similar to the WEEE directive and in line with the EU Green Deal. For the Recovery, and especially for the Circularity scenario, FutuRaM will also consider the effects of proposed ecodesign requirements for sustainable products (e.g., longer lifetimes, increased reusability, repairability, recyclability).

However, for waste that does not consist of discarded consumer products, but instead results from industrial production activities, in particular for MIN and SLASH, we must still produce specific scenarios related to mining, metallurgy, and waste and fuel combustion. The production of new mining wastes will depend on new local mining activity.

Predicted production in the EU until 2050 will be forecast (equally across the three scenarios) and the flows into the MIN waste stream can be calculated with the respective transfer coefficients. The recovery of historical MIN stock, which is a target of the CRM Act, should be modelled differently. It requires a hypothesis about the percentage of historical tailings recoverable by commodity and country.

The scenarios will account for increasing resource use effectiveness and production process efficiency thus indicating lower volumes and quality of generated production residues (both by-products and waste such as red mud, waste rock, slags, etc.) per unit of product (expressed either as product mass or product value), whether that product is a metal (e.g., a copper cathode), metal alloy (e.g., aluminium alloy n° 5183) or metal product (e.g., cold rolled stainless steel sheet).

Excepting the BAU storyline, WEEE, ELV, and BATT waste material recovery will follow the targets in the EU.

For SLASH and MIN, we will evaluate recent trends in waste generation and extract plausible ranges of generation toward 2050.

For CDW, embedded WEEE will follow EU targets, and bulk waste will incorporate storylines and scenarios that are congruent with predicted demolition rates (where renovation is the alternative emphasised in the CIR storyline).

Various drivers will be assigned to move between these ranges and will be key to the specific, harmonized storyline for the scenario. Finally, the targets and storylines will be aligned with assumptions on technology development.

Consideration of geopolitical developments

The storylines also attempt to consider geopolitical considerations and thus supply chain resiliency for satisfying the product demand in the scenarios. We must omit, however, possible changes in waste flow volumes and composition that could arise from any material supply constraints.

The reasoning for this is that it would needlessly confuscate the interpretation of the modelling results as the incertitude of these potentialities is very high and this realm is outside the scope of FutuRaM's mandate and expertise.

The most volatile aspect of the 'criticality calculation' is the risk profile of the producing country. For many material-exporting nations, this is not something that can be reliably forecast, especially not over the next 30 years. Thus, it will be assumed that the growth in material demand for (among other needs) the energy and mobility transitions can be satisfied either by an increase in mining and metallurgy activities within the EU or by growing imports from raw material-producing countries outside the EU.

That is, if we go for increased domestic EU production to minimize geopolitical supply risk, it may indicate more EU production residue generation even under increased production efficiency and resource effectiveness. The increase of domestic industrial activity, as a response to an envisioned increased internal demand, supposes an equivalent rise of societal approval for mining and refining activities on EU territory.

If the increased demand is, however, satisfied by imports from non-EU countries, which we know have domestic resource consumption also growing significantly due to the energy and mobility transition, our assumption would be to shift the mining and refining activities from EU countries towards resource-rich non-EU countries.

This shift would also imply an increased risk for geopolitical instability and/or security of supply of critical raw materials to the EU.

This situation is front of mind for many in policy and business and the EU is 'applying a policy mix that aims to increase domestic capacity, diversify suppliers, and support the multilateral rules-based trade environment.'

However, '…most experts predict that reshoring or nearshoring will be of limited importance. With time, though, resilience may improve through international cooperation, diversification and the accelerated uptake of digital technologies.' [27]

Note: supply constrictions will be considered in the model's sensitivity analysis and the codebase will be designed to allow for the optimisation of the SRM recovery system based on any value statements (such as profit, environmental impacts, supply and demand).

2.2.2. Step 2: Determine methodology

Methodology types and selection criteria

The second step in the scenario development process is to determine the methodology to be used. This involves identifying the most appropriate methods and tools for the specific context and objectives of the scenario development process. The methodology should be selected based on the following criteria:

Relevance:
The methodology should be relevant to the specific context and objectives of the scenario development process.

Applicability:
The methodology should be applicable to the specific context and objectives of the scenario development process.

Feasibility:
The methodology should be feasible given the available resources (e.g., time, budget, expertise, data, etc.).

Transparency:
The methodology should be transparent and well-documented, allowing for replication and further analysis.

Flexibility:
The methodology should be flexible and adaptable, allowing for updates and adjustments as new information becomes available.

Accessibility:
The methodology should be accessible to a wide range of stakeholders, ensuring that it can be understood and used by non-experts.

Effectiveness:
The methodology should be effective in achieving the objectives of the scenario development process.

Efficiency:
The methodology should be efficient in terms of time, cost, and resources required to implement it.

Acceptability:
The methodology should be acceptable to stakeholders, ensuring that it is perceived as fair and legitimate.

Further details are given in this section, and the table in section 13.3 provides an overview of the methods and tools considered, along with a brief description of each and its relevance to the specific context and objectives of the FutuRaM scenario development process.

Choice of methodology

The grant proposal for the FutuRaM project outlined that there should be at least three scenarios developed, namely business as usual, recovery, and circularity. This remains the case; however, during the scenario development process, additional scenarios or scenario dimensions were considered, including supply chain security and the energy transition.

Considered dimension — Supply chain security:

Due to various political developments in 2022, the question of the security of the EU's supply chains for CRMs was brought into focus. This led to the proposal from stakeholders to consider a scenario dimension that would explore the security of the EU's supply chains for CRMs.

Considered dimension — Energy transition:

The energy transition is a key topic in the EU's policy agenda, and the FutuRaM project is concerned with the role of CRMs in the energy transition. Therefore, the proposal was made to consider a scenario dimension that would explore the energy transition in the EU.

Method — Multi-criteria analysis and cross-impact analysis

In order to assess the potential inclusion of these additional scenario dimensions, a multi-criteria analysis and a cross-impact analysis were conducted [96]. The addition of extra dimensions increases the possible number of scenarios significantly. By assessing the consistency and plausibility of these combinations with a matrix-based method, it was possible to reduce the number of scenarios.

For example, low progress in the energy transition is unlikely to concur with high progress in recycling/circularity indicators and can be excluded. In contrast, different levels for the supply chain security dimension would result in an additional scenario, as this dimension is considered independent of the others.

Ultimately, supply chain security was eliminated as a scenario dimension. This is due to the consortium's inability to speculate on geopolitical developments and the added incertitude it would introduce to the scenarios.

The potential of supply constraints will, however, be considered in the future sensitivity analysis of the model, as well as potentially through an array of explorative multi-object optimisation procedures. This can produce projects to answer the question, 'What would happen to the SRM system if element x is constrained, and what would be the optimal response to this constraint?'

Method — Delphi

The Delphi method [97] was used in the initial stages of the scenario-building process to gather and aggregate the opinions of experts or stakeholders. Internal consultation with consortium members who were experts in their respective waste streams or other aspects of the recovery system was conducted.

The method involves steps such as the selection of experts, generation of initial questionnaires, iterative rounds of responses, and convergence and consensus building. For the later stages of the process, further rounds of consultation will be conducted with external stakeholders, including representatives from industry, academia, and government.

Choice of Scenario Type

The general types of scenarios are summarized in Table 2.3. In the context of futures studies, various approaches and methodologies are employed to understand the potential trajectories of future developments [23, 82, 83, 90, 91, 92].

We can classify scenario studies into three primary categories, each addressing distinct questions about the future. These categories are tailored to better align with the specific objectives of scenario usage:

Predictive Scenarios (Answering 'What Will Happen?'):

Pros:
These scenarios offer insights into potential future outcomes, aiding in long-term planning.

Cons:
They are contingent on assumptions and may not account for unexpected events.

Applicability:
Predictive scenarios are valuable when the aim is to forecast future developments under certain conditions.

Explorative Scenarios (Answering 'What Can Happen?'):

Pros:
Explorative scenarios explore a wide range of potential future scenarios, fostering preparedness for various outcomes.

Cons:
They do not prioritize the likelihood or desirability of scenarios.

Applicability:
These scenarios are beneficial when considering multiple potential futures and the need to adapt to diverse outcomes.

Normative Scenarios (Answering 'How Can a Specific Target Be Reached?'):

Pros:
Normative scenarios focus on achieving predefined objectives and offer guidance on strategies to attain them.

Cons:
They are inherently normative, starting with specific goals in mind.

Applicability:
Normative scenarios are suitable when the objective is to work towards predefined targets and develop actionable plans to reach them.

The choice of scenario category is influenced not only by the characteristics of the system under study but also by the user's worldview, perceptions, and study objectives. Additionally, the user's perspective plays a crucial role in determining the most suitable approach. For instance, the decision to employ predictive, explorative, or normative scenarios hinges on the user's goals and the nature of the questions they seek to answer.

Furthermore, considerations regarding the predictability of the future and the potential for influencing it can impact the selection of scenario types. For example, some users may argue that uncertainty in certain parameters makes long-term predictions less meaningful, while others may see value in using forecasting and optimisation models to stimulate discussions and inform decision-making processes.

In practice, a combination of qualitative and quantitative techniques can be employed to create scenarios tailored to specific needs. For instance, a blend of techniques may be used to generate forecasts, especially when external factors are uncertain. Likewise, strategic scenarios often begin with external scenario generation and proceed to identify available policy options.

Table 2.3: Types of scenario (adapted from [23, 82])

SCENARIO CATEGORY	SCENARIO TYPE	OUTCOME	TIMEFRAME	SYSTEM STRUCTURE	FOCUS ON FACTORS
Predictive *what will happen?*	Forecasts	Typically quantitative, sometimes qualitative	Often short	Typically one	Typically external
	What-if	Typically quantitative, sometimes qualitative	Often short	One to several	External and, possibly, internal
Explorative *what can happen?*	External	Typically qualitative, quantitatively possible	Often long	Often several	External
	Strategic	Qualitative and quantitative	Often long	Often several	Internal under influence of the external
Normative	Preserving	Typically quantitative	Often long	One	Both external and internal
	Transforming	Typically qualitative with quantitative elements	Often very long	Changing, can be several	Not applicable

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The scenarios developed in the FutuRaM project are a combination of predictive and normative:

BAU:
What will happen if current trends continue? This scenario is predictive in nature, based on the assumption that the current trends and developments in waste management and resource recovery systems will continue into the future.

Recovery:
What will it take to achieve the EU's targets for material use and recovery?
Focus on technology This scenario is normative, focusing on manipulating the technology and infrastructure of the recovery system to achieve the EU's targets and mandates.

Circularity:
What will it take to achieve the EU's targets for material use and recovery?
Focus on re-X strategies This scenario is a combination of normative and explorative, considering the targets and mandates of the EU's circular economy action plan and exploring re-X strategies in the recovery system.

The methodology and scenario types were selected based on their relevance, applicability, feasibility, transparency, flexibility, accessibility, effectiveness, efficiency, and acceptability to the scenario development process.

2.2.3. Step 3: Marker-scenario mapping

Justification and methodology

This preliminary step in the scenario development process involves conducting a literature study to identify existing scenarios that are relevant to the FutuRaM project. This step is crucial as it serves several important purposes and provides valuable insights for the overall scenario development process. It helps the scenario development team to build on existing knowledge, identify relevant scenarios, gain insights and inspiration, fill knowledge gaps, and enhance credibility and comparability.

Building on existing knowledge
Conducting a literature study allows the FutuRaM project team to tap into existing knowledge and expertise in the fields of waste management, resource recovery, and circular economy. It provides a foundation of existing scenarios that have been developed by other researchers, organizations, or institutions. By building on this existing knowledge, the FutuRaM project can leverage the insights, methodologies, and findings from previous scenario studies, saving time and resources.

Identifying relevant scenarios
Marker scenario mapping helps identify scenarios that are relevant to the specific objectives and scope of the FutuRaM project. By reviewing the literature, the project team can assess the applicability of existing scenarios to their research questions and determine which scenarios align with the waste streams, sectors, and policy domains being considered. This step ensures that the scenarios selected for further analysis are well-suited to address the project's goals.

Gaining insights and inspiration
Reviewing existing scenarios provides the FutuRaM project team with valuable insights and inspiration for the development of their own scenarios. It allows them to understand the different approaches, assumptions,

and methodologies used in previous scenario studies. This knowledge can inform the design and structure of the FutuRaM scenarios, helping to ensure a rigorous and well-founded approach.

Filling knowledge gaps
Marker scenario mapping helps identify any gaps or areas of limited knowledge in the existing scenario landscape. It allows the FutuRaM project team to identify topics or aspects that have not been adequately addressed in previous scenarios. This awareness of knowledge gaps can guide the project team in focusing their efforts on areas where new insights and contributions can be made, leading to a more comprehensive and innovative scenario development process.

Enhancing credibility and comparability
By conducting a literature study and referencing existing scenarios, the FutuRaM project can enhance the credibility and comparability of their own scenarios. The project team can reference and compare their findings, assumptions, and results with those from previous studies, contributing to the overall body of knowledge in the field. This promotes transparency, robustness, and consistency in the scenario development process and allows for better benchmarking and evaluation of the FutuRaM scenario set.

Content of the marker scenario mapping for application to FutuRaM's scenarios

Table 13.4 in section 13.4 presents an overview of the marker scenarios considered in the FutuRaM project. The table is not intended to be exhaustive but rather to provide an overview of the different scenarios that have been developed in the fields of waste management, resource recovery, and circular economy.

2.2.4. Step 4: Identification of key drivers of change

In this step, the key drivers of change that will shape the future of the scenarios are identified. Key drivers are the factors or forces that have a significant influence on the waste management system and its development over time. These drivers can be social, economic, technological, environmental, or policy-related.

The purpose of identifying key drivers of change is to understand the factors that will have the greatest impact on waste management and to ensure that the scenarios capture the range of possible outcomes influenced by these drivers.

The process of identifying key drivers involves a combination of literature review, expert consultations, and stakeholder engagement. It requires a comprehensive analysis of relevant trends, uncertainties, and emerging issues that may affect the waste management system.

The key drivers identified in this step will be used to develop the storyline themes and scenario parameters in the next step.

Figure 2.2 illustrates the process of identifying key drivers of change.

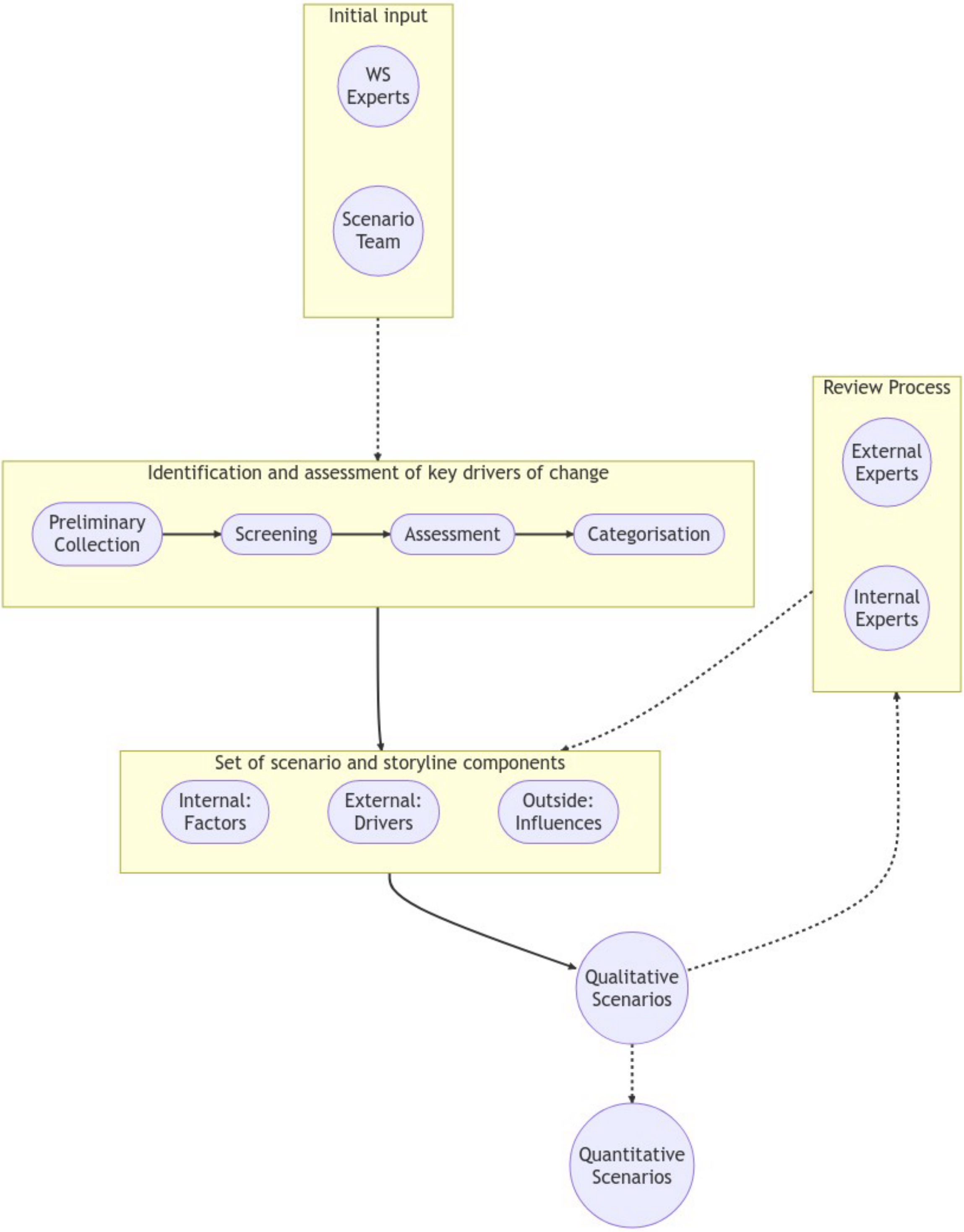

Figure 2.2: An illustration of the process used for identifying key drivers of change

Methodology and results of this stage in FutuRaM's scenario development:

The overall goal of this process is to identify and include elements in the storylines and scenarios that are relevant, plausible, and influential in shaping the future.

The selection, screening, and categorisation steps ensure that the elements chosen for the development of storylines and scenarios are consistent, coherent, and aligned with the objectives and scope of the scenario exercise.

1. **Preliminary collection:**

   This step involved gathering a pool of potential elements that could be included in the storylines and scenarios.

   These elements were derived from expert input from waste streams and the scenario development team, including taking knowledge from the literature review and existing scenarios identified in Step 2 — Marker scenario mapping.

   This step was conducted using the PESTLE analysis framework. The PESTEL (or PESTLE) framework is a strategic tool used to understand the macro-environmental factors that can affect a system.

   A PESTEL analysis can help identify opportunities and threats linked to each of these factors, understand the broader context, and shape scenarios accordingly [98, 77].

   The acronym PESTEL stands for:

   Political:
   These factors refer to the impact of government policies, regulations, and political stability. This includes issues like tax policy, labour laws, environmental regulations, trade restrictions and reforms, tariffs, and political stability.

   Economic:
   These factors relate to the broader economic environment, including factors like economic growth, exchange rates, inflation rates, interest rates, disposable income of consumers and businesses, and the general health of the economy.

   Sociocultural:
   These factors include societal trends and characteristics that could affect your business. They include demographic trends (like age, gender, and ethnicity), cultural trends, lifestyle preferences, consumer attitudes, and broader societal expectations.

   Technological:
   These factors refer to the impact of emerging technologies, research and development activities, automation, the rate of technological change, and the adoption of technology within your market.

   Environmental:
   These factors refer to ecological aspects that can affect a system. This includes environmental regulations, consumer attitudes towards sustainability, climate change, and other natural events.

   Legal:
   These factors include laws and regulations with which your business must comply. These can include labour law, consumer law, health and safety law, and restrictions on the import or export of goods.

   The 68 elements identified in the initial screening stage are listed in section 13.5.

2. **Screening:**

   In the screening step, the collected elements are evaluated and assessed based on specific criteria. This was conducted through a literature study and internal consultation of scientists in the project. This evaluation helps determine the relevance, reliability, and significance of each element for the development of storylines and scenarios. Many elements were aggregated, especially if they were deemed to follow similar trends to others (e.g., recyclability mandates and improved recyclability in project design). Elements that did not meet the predefined criteria or were deemed irrelevant, 'un-modellable' or unreliable were excluded from further consideration (e.g., corruption, data protection, and supply chain conflict).

   The 28 elements that were identified in this stage are listed in section 13.6.

   In Figure 2.3, an excerpt of a spreadsheet illustrates part of the screening process for the FutuRaM scenarios, which was informed by the waste streams. In this exercise, the elements were evaluated based on their relevance to the waste streams and their potential impact on the waste management system.

   The elements were also assessed based on their plausibility and likelihood of occurrence in the future. The elements that were deemed relevant, plausible, and influential were included in the storylines and scenarios.

THIS TABLE IS FOR THE ASSESSMENT OF THE RELEVENCE OF EACH SCENARIO ELEMENT TO INDIVIDUAL WASTE STREAM FLOWS	ELV			BAT					WEEE					
DRIVER/FACTOR	Bulk metals	Critical raw materials	Average	Portable Batteries	Industrial Batteries	Automotive (SLI) Batteries	EV Batteries	Average	CAT-I - Temperature exchanger	CAT-II Screens	CAT-III Lamps	CAT-IVa Large equipments	CAT-IVb PV	CAT-' Smal equi nts
Population				5.00	5.00	4.00	5.00	4.75	5.00	5.00	5.00	5.00	5.00	5.
Resource shortage	3.00	5.00	4.00	5.00	5.00	2.00	5.00	4.25	4.00	5.00	4.00	4.00	5.00	4.
Treatment cost				4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.
Digital product passports	3.00	3.00	3.00	4.00	4.00	4.00	4.00	4.00	3.00	3.00	3.00	3.00	3.00	3.
Obsolescency	1.00	5.00	3.00	4.00	4.00	3.00	4.00	3.75						
Digitalization	1.00	5.00	3.00	4.00	4.00	3.00	4.00	3.75						
SRM prices				4.00	4.00	2.00	4.00	3.50	4.00	4.00	4.00	4.00	4.00	4.
Product prices				3.00	4.00	1.00	4.00	3.00	3.00	5.00	3.00	3.00	3.00	3.
Recyclability mandates	4.00	5.00	4.50	3.00	3.00	3.00	3.00	3.00	2.00	3.00	2.00	2.00	2.00	3.
Conflict in supply chain	4.00	5.00	4.50	4.00	4.00	0.00	4.00	3.00	2.00	3.00	2.00	2.00	3.00	2.
Obligatory recycling standards for treatment facilities				3.00	3.00	3.00	3.00	3.00	2.00	2.00	2.00	1.00	1.00	2.
Improved durability	4.00	5.00	4.50	3.00	3.00	1.00	3.00	2.50						
Composition change				3.00	3.00	0.00	4.00	2.50						
Subsidies				2.00	3.00	1.00	3.00	2.25	3.00	2.00	3.00	4.00	4.00	2.
Availability of recovery technologies				3.00	3.00	0.00	3.00	2.25	1.00	4.00	1.00	1.00	4.00	4.
Taxation (raw materials, landfill)	4.00	4.00	4.00	2.00	2.00	3.00	2.00	2.25	2.00	2.00	2.00	2.00	4.00	2.
Obligatory removal of CRMs from waste				3.00	3.00	0.00	3.00	2.25	1.00	2.00	2.00	1.00	2.00	2.
Corruption	2.00	2.00	2.00	3.00	3.00	0.00	3.00	2.25	1.00	1.00	1.00	1.00	1.00	1.
Supply chain due diligence laws	4.00	4.00	4.00	0.00	4.00	0.00	4.00	2.00	0.00	1.00	0.00	0.00	1.00	1.
Improved recyclability	4.00	5.00	4.50	2.00	2.00	0.00	2.00	1.50						
Ecodesign				2.00	2.00	0.00	2.00	1.50						
Trade barriers	3.00	5.00	4.00	2.00	2.00	0.00	2.00	1.50	2.00	3.00	2.00	2.00	3.00	2.
Industrialisation of Europe	4.00	5.00	4.50	0.00	2.00	0.00	3.00	1.25	3.00	3.00	1.00	3.00	3.00	1.
Reduced consumerism	5.00	3.00	4.00	0.00	1.00	4.00	0.00	1.25	1.00	3.00	2.00	1.00	0.00	2.
Accessibility/Infrastructure			#DIV/0!	3.00	0.00	0.00	0.00	0.75	3.00	4.00	4.00	3.00	3.00	4.
New mines in rich EU countries?	3.00	5.00	4.00	1.00	1.00	0.00	1.00	0.75	3.00	2.00	3.00	4.00	4.00	2.
Miniturisation	3.00	5.00	4.00	1.00	0.00	0.00	0.00	0.25						
Sharing economy	4.00	4.00	4.00	1.00	0.00	0.00	0.00	0.25	1.00	1.00	1.00	3.00	1.00	1.
Repairability mandates	5.00	5.00	5.00	0.00	0.00	0.00	0.00	0.00	2.00	3.00	3.00	3.00	2.00	3.
Renewable energy targets				0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00	0.

Figure 2.3: An excerpt of a spreadsheet used as part of the screening process

3. **Assessment:**

   Once the screening process was complete, the remaining elements were aggregated and categorized based on their thematic relevance or characteristics. This categorisation helps organize the elements into meaningful groups or themes that align with the objectives and scope of the scenarios. The 21 elements that were identified in this stage are listed in Table 2.4. Note that CIR and REC are very similar for many scenario elements, the main difference being the way in which the targets are achieved. That is, for CIR, re-X strategies are promoted, whereas, for REC, the focus is on technological advancements in the recovery system. This distinction will have a significant impact on how the scenarios are quantitatively modelled and on the subsequent outcomes of these models.

Table 2.4: List of drivers and factors identified in the screening phase

DOMAIN	DRIVER/FACTOR	DEFINITION	INTERNAL	BAU	REC	CIR
TECH	Recovery technology	Implementation and advancements in waste recovery technologies	TRUE	I	III	III
TECH	Product technology	Changes in product function or composition	TRUE	I	III	III
TECH	Integration of SRM system across EU	Integration of a secondary raw material recovery system across EU countries	TRUE	I	III	III
ENV	Increased drive for environmental protection	Growing concern and motivation for environmental conservation	TRUE	I	III	III
ECO	Progress toward renewable energy targets	Advancements and achievements in renewable energy generation	TRUE	III	III	III
ECO	Subsidies and taxation to promote circularity	Financial incentives or taxes to encourage circular economy	TRUE	I	II	III
SOC	Participation in re-X activities	Engagement in refuse-reduce-repair-reuse activities	TRUE	I	I	III
POL	Stricter environmental regulations	Tightening of environmental laws and regulations	TRUE	II	III	III
POL	Stricter waste management regulations	Strengthening of waste management laws and regulations	TRUE	II	III	III
POL	Supply chain due diligence laws: implementation and enforcement	Obligations for identifying and mitigating negative impacts in supply chains	TRUE	I	III	III
POL	Compliance with waste targets	Meeting specific waste management and recycling targets	TRUE	I	III	III
ENV	Resource shortages	Limited availability of natural resources	FALSE	n/a	n/a	n/a
ECO	Raw material vs SRM prices	Price dynamics and competition between raw materials and secondary raw materials	FALSE	n/a	n/a	n/a
ENV	Climate change impactsmitigation	Effects and actions related to climate change	FALSE	n/a	n/a	n/a
ECO	International trade and co-operation (vs. autarky)	Collaborative trade agreements and global cooperation	FALSE	n/a	n/a	n/a
ECO	Energy prices	Costs and fluctuations in energy prices	FALSE	n/a	n/a	n/a
ECO	Economic growth	Overall economic expansion and development	FALSE	n/a	n/a	n/a
ECO	Re-industrialisation of EU	Shift towards increased industrial activities in the EU	FALSE	n/a	n/a	n/a
SOC	NIMBY to projects	Opposition to local projects and developments	FALSE	n/a	n/a	n/a
SOC	Population and urbanisation	Growth and urban development of population	FALSE	n/a	n/a	n/a
ECO	CO2 market price	Price and market dynamics of carbon emissions	FALSE	n/a	n/a	n/a

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4. **Categorisation**

The scenario elements were then assessed based on their potential impact on the waste management system. For each element, an assessment was made as to whether it was within the scope of FutuRaM to include them as variables in the models, and therefore also the scenarios and their storylines.

Those deemed to be within the scope are 'internal' and will be intensively researched and modelled (e.g., composition and design changes).

Those deemed to be outside the scope are 'external' and will be included in the storylines, will vary over time, but will not vary across the three scenarios (e.g., population and GPD).

Those deemed to be outside the scope and also outside the influence of the waste management system are 'outside' and will not be included in the storylines or scenarios, though, in some cases, may be considered in the sensitivity analysis (e.g., supply constraints).

Justification for keeping certain elements outside of the scenario models:

The purpose of the FutuRaM project is not to provide all-encompassing scenarios that attempt to capture every possible future development. Such scenarios are inherently inaccurate and can give a false sense of certainty to the model's outcomes. Instead, the focus of FutuRaM is specifically on the Sustainable Resource Management (SRM) system and its implications for the future. Therefore, the scenarios developed within FutuRaM should selectively incorporate elements that have a direct impact on the SRM system.

Furthermore, the scenarios should prioritize elements that can be considered as 'policy knobs', meaning variables or factors that can be adjusted or controlled to test different settings. By including these, the scenarios can explore the effects of different policy decisions or interventions on the SRM system's outcomes. This targeted approach ensures that the scenarios generated are relevant to the project's objectives and facilitate meaningful analysis.

It is crucial to avoid excessive complexity and convolution in scenario modelling. When there are too many convoluted elements included, the results of the modelling exercise can become, at best, difficult to understand and interpret. At worst, the outcomes may become practically useless due to the overwhelming interactions and uncertainties introduced by the complex elements. Therefore, careful consideration is necessary to strike a balance between incorporating essential factors and maintaining the clarity and usefulness of the scenario modelling results.

Examples:

Resource shortages:

Resource shortages can be highly unpredictable and subject to various external factors such as geopolitical events, natural disasters, or technological advancements. The precise timing and extent of resource shortages are challenging to forecast accurately, making it difficult to include them within the model without introducing significant uncertainty.

This is especially true for the long-term time horizon of the FutuRaM scenario set. This factor will, however, be considered in the sensitivity analysis of the model and additionally, the codebase will be designed to allow for the optimization of the SRM recovery system based on any supply-demand value statements.

Raw material vs SRM prices:

The dynamics and competition between raw materials and secondary raw materials can be complex and influenced by various market factors, technological advancements and policy interventions. As with resource shortages, these dynamics are challenging to forecast accurately, making it difficult to include them within the model without introducing significant uncertainty.

It will, however, be possible to couple the model with a market model to explore the effects of different price dynamics on the SRM system's outcomes. This could be considered in a multi-objective optimization procedure performed as an extension to the model.

2.2.5. Step 5: Develop storyline themes

Given that the scenario themes and directions were broadly dictated by the FutuRaM project charter, the rough shapes of the storyline narratives were already defined. That is: the effects on the availability of SRMs from the development of the SRM recovery system and the development of re-X strategies.

2.2.6. Step 6: Qualitative narrative development

The scenario storylines will be described in detail in the next section. This step involved taking the themes defined by the charter and the elements identified in the previous steps and working with the internal waste stream groups to develop qualitative estimates about how each of these elements (at their different levels) may have an impact on the amounts and composition of the SRM flows in their purview.

2.2.7. Step 7: Definition of scenario parameters

The scenario parameters are the set of quantitative values or functions that will be used to define the scenario inputs for the model. Details of these parameters can be found in chapter 4.

2.2.8. Step 8: Quantitative modelling

Full details of the scenario quantification process can be found in chapter 4.

2.2.9. Step 9: Implementation

The scenario implementation will be performed in the next stages of the project.

2.2.10. Step 10: Review process

The review process is intended to ensure that the elements included in the storylines and scenarios are relevant, plausible, and consistent with the scenario objectives and scope.

The first stage of the review process is to open the scenario development process to the wider FutuRaM consortium. This will be done by sharing the scenario development process and the results of the assessment and categorisation step with the consortium and inviting feedback and suggestions. The feedback will be used to refine the elements and their categorisation and to identify any elements that may have been missed in the initial assessment.

The second stage will involve opening the scenario development process to external stakeholders and subject matter experts.

The scenario review process will be performed repeatedly over all stages of the project. This document is a living document and will be updated as the project progresses.

Conclusion of methodology section

The methodology used for the FutuRaM scenario development ensured that the selected elements were relevant, plausible, and influential. The use of the PESTEL analysis framework and Delphi method during the preliminary collection phase provided a comprehensive overview of the macro-environmental factors.

Furthermore, the screening process and the assessment by internal experts ensured that the selected elements were coherent, consistent, and aligned with the objectives and scope of the scenario exercise.

The final list of scenario elements is suited to the goal of the FutuRaM project — to quantify the future availability of SRMs and to evaluate EU material autonomy — and will be used to develop the three FutuRaM scenarios into a quantitative model.

Scenario storylines

CHAPTER CONTENTS

3.1. SCENARIO I: BUSINESS-AS-USUAL

3.1.1. Context

> **Can the Supply of Natural Resources Really Be Infinite? Yes!**
> Hold your hat — our supplies of natural resources are not finite in any economic sense. Nor does past experience give reason to expect natural resources to become more scarce. Rather, if history is any guide, **natural resources will progressively become less costly, hence less scarce**, and will constitute a smaller proportion of our expenses in future years. Population growth is likely to have a long-run beneficial impact on the natural-resource situation.
> *— Julian Lincoln Simon in The Ultimate Resource [70, 71]*

> Population growth, along with over-consumption per capita, is driving civilisation over the edge: billions of people are now hungry or micronutrient malnourished, and climate disruption is killing people. [Societal collapse] is a near certainty in the next few decades, and the risk is increasing continually as long as perpetual growth of the human enterprise remains the goal of economic and political systems. As I've said many times, **'perpetual growth is the creed of the cancer cell'**
> *— Paul Ehrlich [155], see also [72, 73, 99]*

3.1.2. Storyline narrative

This scenario envisions the future based on the current situation, extending to 2050 with very little deviation from present consumption patterns and the secondary raw material (SRM) system [28]. While there may be advances in some areas such as resource efficiency, recovery technology, and the energy transition, substantial modifications remain hindered by economic, social, and political constraints. The primary extraction of raw materials continues to be the primary source to meet the EU's demand.

In the Business As Usual (BAU) scenario, we are projecting the trajectory of the present into the future, extending up to the mid-century mark, 2050, with minimal disruption to existing consumption habits and the secondary raw material (SRM) system. This scenario unfolds on the assumption that the current pace and direction of technological, economic, and social development continue unhindered, and is characterised by a strong persistence of today's patterns.

In this scenario, we see moderate improvements in resource efficiency, advancements in recovery technology, and a slow transition towards greener energy sources. However, these developments are only minor tweaks to the existing system, failing to disrupt or fundamentally alter the established structure. The potential for transformational change remains largely untapped due to various hurdles. Economic constraints, social resistance to change, political inertia, and entrenched interests act as barriers to change, stifling efforts toward a more sustainable SRM system.

Primary extraction of raw materials remains the dominant source for raw materials consumed in the EU, continuing the linear 'take-make-dispose' model of resource consumption (see Figure 3.1). Base metals are well recycled, given their developed markets and economies of scale but rare/special metals are wasted because recycling technologies and economics do not allow for their recovery. Recycling and recovery rates remain stubbornly low, resulting in significant CRM waste. Meanwhile, material demand continues to rise in tandem with GDP growth, further exacerbating the resource pressure.

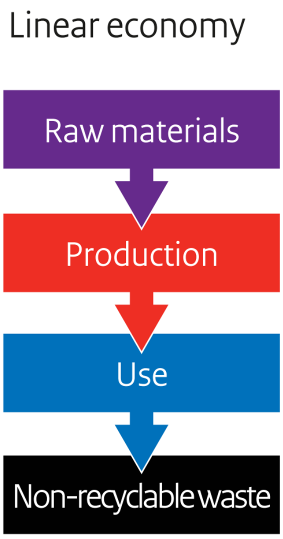

Figure 3.1: The linear economy model in the business-as-usual scenario [9]

Moreover, the environmental impacts of mining and extraction persist as a significant concern. These operations continue to degrade ecosystems, leading to loss of biodiversity and contributing to climate change [29]. Simultaneously, the EU becomes increasingly dependent on imports of SRMs, raising concerns about supply chain security and geopolitical risks [29].

Innovation in SRM recovery technologies is hampered by a lack of investment and regulatory support. The focus remains predominantly on cost-effective material production and use, with little regard for environmental implications or long-term sustainability. Material scarcity and price fluctuations, therefore, may become a considerable risk to the EU industry, limiting stable penetration of new recovery technology and threatening economic stability.

Moreover, the tightening of environmental regulations is restricted, inadequately addressing emerging challenges or incentivising sustainable practices. The lack of regulatory progress may further exacerbate environmental damage and biodiversity loss.

In essence, the BAU scenario is characterised by a continuation of current trends and practices, a future where the potential for a sustainable SRM system is unrealised due to the stranglehold of prevailing economic, social, and political constraints.

In the Business-as-usual (linear economy) scenario, the following are key characteristics:

- A forecasting model is used to predict the future based on the current situation and the development of existing trends.
- Many EU targets for recycling and recovery are not met, and the current linear model largely persists.
- Material demand keeps pace with GDP growth, perpetuating a trend of increasing consumption. Primary mining and extraction persist as the leading sources of raw materials, underlining the dependency on traditional extraction methods.
- Recycling and recovery rates continue to lag, leading to an accumulation of SRM waste that signals missed opportunities for resource reuse.

- The environmental repercussions of mining and extraction, such as land degradation and water pollution, continue to be a pressing concern, reflecting the ecological toll of this linear model.
- The EU's dependency on imports of SRMs escalates, heightening the risk of supply disruptions. While supply disruption can serve to stimulate investment in new SRM recovery, volatility stifles innovation and advancements in this field.
- The industrial focus remains on cost-effective material production and use, disregarding the long-term sustainability aspect.

3.1.3. Waste stream specific scenario impacts

BATT (Battery waste)

Sources: [24, 86, 7, 8]

In the business-as-usual (BAU) scenario, the management of end-of-life batteries remains largely unchanged. The lack of technological innovation and regulatory incentives leads to a continued low recovery rate of valuable materials from battery waste.

- A growing volume of battery waste due to the increased use of electronic transport and renewable energy storage systems.
- Lack of technological innovation and regulatory incentives lead to low recovery rates for certain battery types and certain elements.
- Collection systems for battery waste remain sporadic and unstandardised.
- Primary extraction remains the dominant source of battery materials.
- Share of LIB will increase (EV, LMT, Industrial LIB uptake)
- LIB Battery Chemistries will change and new LIB technologies will enter the market. Though, not with a focus on recycling and recovery.
- Larger portable batteries: shift towards Li-ion batteries
- Small format batteries in EEE: no significant change in battery chemistry.
- Use of critical resources continues but is already decreasing (BATT chemistry already changing towards less CRM content)
- Large-scale reuse of batteries is minimal
- Collection rates do not fulfil the EU targets
- Recycling efficiencies do not fulfil the EU targets
- Recovery rates do not fulfil the EU targets

ELV (End-of-Life Vehicles)

Sources: [86, 142, 30, 31, 32, 33]

The BAU scenario maintains the current approach to end-of-life vehicles, with minimal improvements in the recovery and recycling process. The absence of effective technologies and regulatory incentives results in low recovery rates of valuable materials from ELVs.

- Legislation banning new ICEVs from 2035
- Current recovery technologies are unable to significantly improve the extraction of valuable materials from ELVs.
- Consumer demand continues to drive high production of new vehicles.
- ELV collection systems remain at their current efficiency.
- A significant proportion of vehicle components continue to end up as waste.
- Gradual and slow improvement of recycling chain technology efficiency
- No new legislation to improve recovery and support circular strategies in comparison to 2023

WEEE (Waste Electrical and Electronic Equipment)

Sources: [5, 6, 34, 35, 36, 78]

In the BAU scenario, the treatment of WEEE does not significantly change. The lack of technological progress and effective regulation results in low recovery rates of valuable materials from WEEE.

- Limited improvements in the recovery of valuable materials from WEEE.
- High consumer demand for new electronics continues to drive high WEEE generation.
- Ineffective collection systems and lack of public interest result in significant amounts of WEEE ending up in landfills.
- No significant growth in collaboration between government and industry for WEEE recovery.
- The majority of WEEE continues to be treated with common domestic waste, with low recycling rates.
- No groundbreaking technologies and practices to improve recovery and circularity.
- Reuse of products and components is not widely utilised
- Changes in legislation (e.g., circular economy and product design targets, targets for collection and recycling) are not strictly implemented.
- The BAU and the REC scenarios are similar from the put-on-market perspective (e.g., production and consumption remain the same), but it‘s the recovery stage that makes the difference.

MIN (Mining Waste)

Sources:

The BAU scenario sees the continuation of current practices in mining waste management. The absence of advanced recovery technologies and regulatory incentives leads to low recovery rates of valuable materials from mining waste.

- Limited technological advancements lead to static recovery rates of valuable materials from mining waste.
- Continued reliance on primary extraction as the dominant source of raw materials.
- Minimal advances in collaboration between government and industry for mining waste recovery.
- Low levels of traceability and management of mining waste.
- Mining waste remains a significant environmental challenge.
- Mining waste recovery projects remain too expensive.
- Little incentive for the private sector and public sector, except for monitoring environmental risks of existing deposits.

CDW (Construction and Demolition Waste)

Sources: [10]

In the BAU scenario, the management of Construction and Demolition Waste (CDW) remains largely unchanged.

- Focus on new construction to meet demand, no changes in CDW generation rate.
- No increase or refurbishment or renovation activities relative to new construction rates.
- Continue meeting the 2020 EU target from the Waste Directive [10] of 70% CDW recovery (including preparation for re-use, recycling, and other material recovery, including backfilling)
- Recovery of metals remains on already high levels (90%) [100].
- Recovery of minerals remains on already high levels (70%) by using them as aggregates in road construction and backfilling [100].
- Recycling of wind turbines stays around 85% (mainly metals), permanent magnets continue to be recycled as part of the metal fractions.[CITATION]
- Base metals are recovered as they have been, though there are limited improvements in recovery technologies and regulatory incentives.
- Repowering trends for wind turbines persist.
- Excluding wind turbines, there is no particular focus on the recovery of CRMs from CDW, where they constitute only a small fraction of the total mass (e.g., embedded in scrap steel).

SLASH (Slags and Ashes)

Sources: [143, 144, 101, 145, 102]

In the BAU scenario, SLASH continues to be treated generally as low or negative-value waste. The absence of economically profitable recovery technologies or regulatory mandates leads to low improvements in the recovery rates of CRMs from SLASH.

SLAGS

Slags are waste products from the metallurgical industry and contain mainly minerals with some metals that could not be recovered during the metallurgical process. The reasons the metals are not recovered are:

1. Current technology does not allow further recovery
2. The metals are not of economic interest.

More than 90% of the slags are minerals, therefore the slags that are in line with the environmental criteria (end of waste criteria) are valorised as aggregates for the construction industry or as SCM (cement replacement materials).

Slags containing high concentrations of heavy metals are landfilled.

- The volume of slags will stay stable. From 2013-2023, there was a stable production of metals, which results in a stable production of slags.
- Due to the energy transition, it is expected the volume of slags from "classic" furnaces (eg BOF) will reduce and the number of slags from electric arc furnace will go up.
- There are limited facilities to recover CRMs/SRMs, but the metallurgical companies do their best to recover as much of the metals as per their economic interests.

Ashes

Ashes are the waste products from the incineration of fossil fuels, biomass, or waste. We distinguish two main types: fly ashes and bottom ashes. Coal fly ashes are used as SCM to replace cement up to 30%. Biomass (bottom and fly) ashes are used (based on the composition) as fertilizer or landfilled. Fly ashes from waste incineration are landfilled, while the bottom ashes are — depending on the location in the EU — further treated.

The fractions that are rich in ferrous metals, aluminium, copper, and zinc are separated and used as input materials for the Fe, Cu, Zn, Al industries. By processing the Cu-rich fractions, PGMs, Ag, and Au are also recovered. The main part of the bottom ashes consists of mineral aggregates, which are used in the construction industry (when they are in line with the environmental criteria). In case the ashes are not treated (or only partly treated), they will be landfilled.

- In the last 10–20 years in some EU countries, we have seen a shift from landfilling towards incineration. This will lead to an increase in the volume of ashes from waste incineration.
- There was a drop in the use of coal as a fuel source, so the volume of coal ashes dropped over the last 10 years and is expected to drop further.
- At the same time the volume of biomass incineration went up. It is uncertain if this volume will increase in the future further (due to the high pressure on land that is needed to produce this biomass).
- Almost all coal fly ashes are used as SCM (high-value stream up to 80 EUR/ton). For Biomass fly ashes there are no sorting facilities, while for waste inceration bottom ashes, there are sorting installations in place, but for the moment, they are not present in all EU countries, and therefore, there is still room for improvement.

3.2. SCENARIO II: RECOVERY

3.2.1. Context

> We are in a period of economic transition. The 'cowboy economy' of the past is obsolescent, if not obsolete. Environmental services are no longer free goods, and this fact is driving major changes. Recycling is the wave of the (immediate) future. **The potential savings in terms of energy and capital have long been obvious. The savings in terms of reduced environmental impact are less obvious but increasingly important.** The obstacle to greater use recycling has been the fact that economies of scale still favor large primary mining and smelting complexes over (necessarily) smaller and less centralized recyclers. But this advantage is declining over time as the inventory of potentially recyclable metals in industrialized society grows to the point that efficient collection and logistic systems, and efficient markets, justify significant investments in recycling. **Increasing energy and other resource costs, together with increasing costs of waste treatment and disposal, will favor this shift in any case.** But government policies, driven by unemployment and environmental concerns, taken together, may accelerate the shift by gradually reducing taxes on labor and increasing taxes on extractive resource use.
>
> *— Robert U. Ayres [103]*

> One does not require much of an imagination to picture how complex it is to physically and chemically separate these components again to ultimately retrieve the metals. **It's just about as difficult as recycling your morning cup of coffee into its ingredients…** There are no simple answers here — at some point, the amount of effort also outweighs the value of the metal content. This knowledge should prompt a consciousness shift in our utilization of our limited resources…
> **The inconvenient truth is that closing the loop is impossible!** Therefore, an honest discussion involves speaking transparently about losses in the process: in the form of energy, metals, and dust, for example. There are technological and economic limits to closing the loop.
>
> *— Prof. Dr. Dr. h.c. mult. Markus Reuter [156]*

3.2.2. Storyline narrative

In the recovery scenario, the central emphasis is on harnessing sophisticated technologies to salvage SRMs from waste streams at the end of their lifecycle. While there are noticeable strides towards the incorporation of 'circular design' principles and re-X strategies, they are mostly seen at the end-of-life and material demand is akin to that observed in the BAU scenario. This is, however, mitigated by the implementation of a comprehensive material recovery system. Figure 3.2 presents a simplified visual depiction of the dominant material flow paths in the recovery scenario.

In this scenario, the central actor is the waste treatment sector, with the spotlight falling on the enhancement of recovery technology. The implementation and optimisation of cutting-edge technologies, such

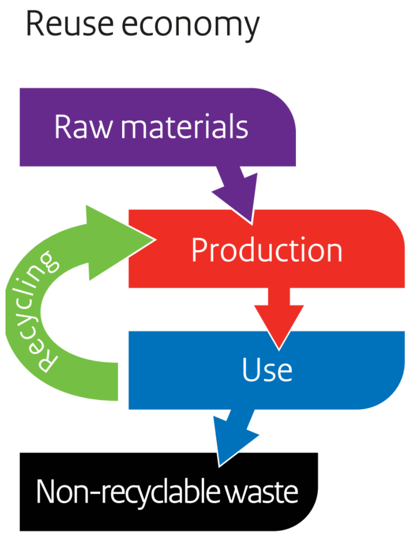

Figure 3.2: The reuse economy model in the recovery scenario [9]

as Artificial Intelligence (AI), automation, and advanced robotics, play a significant role in revolutionising waste treatment processes. These technologies streamline waste sorting, improve the quality of recovered materials, and increase the overall efficiency of the recovery process.

This scenario calls for an emphasis on policy development and standardisation to foster EU-wide development, integration, and compliance. Here, the role of governments and policy-makers becomes crucial in setting more ambitious recovery targets, developing conducive regulatory frameworks, and enforcing compliance. This multi-pronged approach also involves strengthening cross-border cooperation, harmonising waste management standards, and promoting knowledge and technology transfer among EU member states.

To realise more ambitious environmental impact reduction targets, significant progress needs to be made in both technological and policy aspects. Enhancing technological capabilities will improve recovery rates, while robust policy measures will ensure these advancements are integrated into the wider economy in a regulated manner. The future of this scenario depends on the successful fusion of advanced technology, regulatory harmonisation, and a commitment to continuous improvement in waste management and SRM recovery.

Key characteristics of this technology-promoted recovery scenario include:

- This scenario uses a combination of forecasting and backcasting methods to envision the future.
- The backcasting method is used for scenario factors that are covered by governmental targets, starting with the desired outcome and working backwards to the present.
- The forecasting method is used for scenario factors that are not covered by governmental targets, starting with the current situation and extending to the future.
- EU targets for recycling and recovery are met, due to the EU's waste management system becoming more expansive, efficient and effective.
- Technological innovation drives increased recovery rates of SRMs, enabling the more efficient use of waste.
- Digitalisation and automation are more extensively used in recycling processes, leading to enhanced productivity and accuracy.
- There is greater exploration and exploitation of alternative sources such as urban mining, waste streams, and tailings, presenting novel opportunities for resource acquisition.

- New waste regulations and guidelines for SRM recovery are implemented, enforcing better management and extraction of SRMs.
- Investment in research and development for SRM recovery technologies experiences an upswing, promoting continuous innovation in this field.
- Closer collaboration and information sharing between industry and government institutions streamline processes and expedite decision-making.
- New jobs are created in the recycling and recovery sector, offering economic benefits and improving overall employment rates.
- SRM production and use become more efficient and cost-effective, fostering economic sustainability.
- Environmental impact from mining and extraction is reduced, signalling a more sustainable approach to resource acquisition.
- The EU's dependence on primary extraction is reduced, with SRM recovery becoming a more significant source of raw materials.

3.2.3. Waste stream specific scenario impacts

BATT (Battery waste)

Sources: [24, 86, 7, 8]

Under the recovery scenario, end-of-life batteries become a crucial source of secondary raw materials, primarily due to the increased adoption of electric vehicles and renewable energy storage systems. Technological innovation drives the recovery and recycling process, ensuring valuable materials are extracted from waste batteries for reuse.

- Increase in end-of-life batteries due to the growth of electric vehicles and renewable energy storage.
- Advanced recovery technologies facilitate the efficient extraction of valuable materials from battery waste.
- Standardised collection systems enhance the quantity and quality of battery waste available for recovery.
- Industry and government collaboration lead to investments in research and development of battery recovery technologies.
- Battery passports have a strong impact on collection, material recovery rates and recycling rates.
- Collection
  - Portable battery collection increases according to the trend seen in the WEEE waste stream.
  - Improved collection of light means of transport (LMT) batteries.
  - Improved regulation and collection of Industrial batteries.
- Material recovery
  - Improved recycling technologies
  - Battery Pass will improve material recovery
  - Higher recovery rate for lithium
  - Increase in recycling by average weight
  - Recycling of plastics
- Ambitious goals of recycling/recovery rates compete with reuse, so reuse remains low.
- Improved public awareness means that fewer batteries end up in the municipal waste stream and there is less hoarding.
- Against this: there is competition for the batteries from the reuse vs. recycling market.
- Design for recycling (DFR):
  - Material and composition selection for recycling [86].
  - Higher requirements on disassemblability.
  - Information available to promote efficient recovery.

ELV (End-of-Life Vehicles)

Sources: [86, 142, 30, 31, 32, 33]

The recovery scenario envisions a more effective and technology-driven end-of-life vehicle treatment process. Advancements in recovery technologies allow for improved extraction of valuable materials from vehicles at their end of life, although consumerism still drives high demand for new vehicles.

- Innovations in recovery technologies allow for a higher recovery rate of CRM-containing materials from ELVs.
- The total number of vehicles produced remains high due to consumer demand.
- Improved systems for ELV collection are established, ensuring efficient management of ELV waste.
- Increased collaboration between the government and industry leads to investments in ELV recovery technologies.
- Focus on managing end-of-life of vehicles
- EU recovery targets are reached (currently implemented/proposed targets, but also increased and new targets)
- Common/bulk materials (Fe, Non-Fe, plastics etc.,) and precious metals (Au, Ag, Pd, Pt) reach high mass recycling rates and high element recycling rates. Other CRMs currently not recovered reach a moderate level of recovery.
- For instance,
  - More advanced dismantling and processing steps (e.g., components and materials)
  - More specialised recovery of certain components and materials (e.g., electric motors including permanent magnets and embedded REE) as suggested in the proposal for a revised ELV directive.
  - More public and private interest in developing recycling chains
  - Increase in collection rate due to increase in participation from the public and businesses, i.e., target-based incentives with strong regulations and monitoring
- Design for recycling (DFR):
  - Higher requirements on 'disassemblability'.
  - Information available to enable recovery.

WEEE (Waste Electrical and Electronic Equipment)

Sources: [5, 6, 34, 35, 36, 78]

Under the recovery scenario, WEEE becomes a significant resource for secondary raw materials. Technological advancements in the sector improve the efficiency of WEEE treatment, although the consumerism-driven demand for new electronics remains high.

- Advanced technologies enable higher recovery rates of valuable materials from WEEE
- Despite advancements in design for recyclability, WEEE generation remains high due to the consumer demand for new electronics
- Standardised and segregated collection systems for WEEE are implemented, improving the supply of materials for recovery
- Increased industry-government collaboration leads to further development in WEEE recovery technologies
- Consumer behaviour remains a significant hurdle for more efficient WEEE management

- Higher recycling rate — make full use of the disposed parts. For instance:
  - more automation of the dismantling and processing steps (e.g., AI)
  - recycling technologies improvements (e.g., small components recovery is also happening)
  - more effective collection infrastructure
  - financial support provided to recyclers/operators
  - bans on WEEE exports push for increased domestic recycling [37]
- 'Design for recovery' principle — Ecodesign mandates changes in weight and composition of EEE so complexity and the type of materials used
- Higher public awareness and participation on WEEE issue and management
- Higher compliance from the public, the producers and the businesses
- Strong regulations and monitoring are in place with higher collection and recycling targets which are set and implemented and fines are set for those who fail to achieve the targets
- Focus is given more to the EoL management of WEEE

MIN (Mining Waste)

Sources:

Under the recovery scenario, technological advancements enable the extraction of residual valuable materials from mining waste, transforming it into a more valuable resource.

- Technological advancements facilitate the extraction of valuable materials from mining waste.
- Despite progress in recovery technologies, primary extraction remains the dominant source of raw materials due to high consumer demand.
- Government and industry collaboration support the development of technologies for the recovery of materials from mining waste.
- Increased traceability and management of mining waste through digitalisation.
- Mining waste remains a significant environmental challenge.

CDW (Construction and Demolition Waste)

Sources: [10]

Under the recovery scenario, Construction and Demolition Waste (CDW) becomes an important resource for secondary raw materials, though mostly base metals and aggregates. Despite some progress in eco-design and material efficiency, the construction industry continues to generate significant amounts of waste or 'downcycled' materials. Some progress in eco-design and material efficiency, but the construction industry continues to generate significant amounts of waste or 'downcycled' materials.

- Focus on new construction to meet demand, no changes in CDW generation rate.
- No increase or refurbishment or renovation activities.
- Enhancement of the quality of recycling to recover materials at higher value.
- Increased investment and enhanced regulatory system in waste management, contributing to increased recovery.

- Creation of new waste recovery infrastructure that improves recovery.
- Widespread application of selective demolition and strict on-site waste sorting leasing to an increase in recovery of waste.
- Recovery of minerals is intensified with a stronger focus on closed-loop recycling (e.g., cement and aggregate are separated, aggregate is used, but cement is not treated).
- Recovery of other materials like glass, plastics, and wood is also intensified.
- Better separation of waste at source leads to a higher quality of secondary raw materials.
- Repowering trends for wind turbines stay the same.
- Improved recycling of wind turbine blades is notable, especially regarding plastics; permanent magnets are recycled at a functional level.

SLASH (Slags and Ashes)

Sources: [143, 144, 101, 145, 102]

In the recovery scenario, SLASH are recognized as a potential resource for secondary raw materials. Advances in recovery technologies enable the extraction of valuable metals from SLASH, however, the total volume of CRMs recovered from this material remains low, except in cases of supply constraint.

- Digital solutions enhance the traceability and management of SLASH.
- More functional collection infrastructure.
- Financial support provided to recyclers/operators.
- Introduction of SRM/CRM recovery targets. For example, recovery of P from biomass ash for fertilizer.
- Higher awareness and participation of relevant sectors on SLASH issues and management.
- Strong regulations and monitoring are in place with higher collection and recycling targets.

SLAGS

- Advanced recovery technologies allow for the extraction of valuable metals and minerals from slags.
- New recovery technologies are installed in the metallurgical industry.
- All metals are recovered from the slags, and the slag itself only contains minerals or metals as trace elements.
- Due to the low metal content, the slags are ideal resources for the construction industry.

ASHES

- New recovery technologies are installed in the incineration industry.
- We expect a shift in the volume and quality of the ashes.
- Coal fly ashes are the same as in BAU reducing over time to almost zero.
- Biomass ashes have the same volume and quality.

- Waste incineration ashes: due to the recovery scenario there will be a shift from landfill and incineration towards recycling, reuse, and repair. In countries with a high landfill rate, this will lead to higher volumes of ashes. In countries with a low landfill rate, and a high volume of incineration: this will lead to lower volumes of ashes, and a shift in the composition (with less CRM content in the ashes, due to better pre-sorting)

3.3. SCENARIO III: CIRCULARITY

3.3.1. Context

> A circular economy is one that is regenerative by design and aims to keep products, components, and materials at their highest utility and value at all times, distinguishing between technical and biological cycles. This new economic model seeks to ultimately decouple global economic development from finite resource consumption.
>
> *— Ellen MacArthur Foundation [38]*

> **A Circular Economy in the Netherlands by 2050!**
> Imagine we are in the year 2050. In the Netherlands we are living within the planetary boundaries of the Earth rather than permanently crossing them. This is because our relationship with nature, which used to be highly disturbed and led to climate change, has changed for the better …
> … The Netherlands now has a circular economy.
> … The Dutch Government intends to achieve a fully circular economy by 2050 and to halve the use of primary abiotic raw materials by 2030.
>
> *— National Circular Economy Programme, Government of the Netherlands [11]*

> **Circularity of the Dutch economy has barely increased**
> Of all natural resources that were deployed in the Dutch economy in 2020, 13 percent consisted of recycled materials. This percentage is virtually the same as in 2014. In terms of recycling, the circularity of the Dutch economy has barely increased.
>
> *— CBS (Statistics Netherlands) [146]*

> The circular economy promises radical technological transformations in a couple of decades, "nothing less than to open up new and immense horizons for industry", "provide multiple value creation mechanisms", produce "better welfare, GDP, and employment outcomes" [38]. Looking at these claims, one wonders whether policymakers really believe that in 30 years EU citizens will live in an inclusive economy with zero waste, zero emissions, with a perpetual economic growth, continuously absorbing massive flows of immigrants while protecting and enhancing the ecological processes and environmental biodiversity.
>
> …for a young scientist it is not advisable and even shocking to say in policy circles that:
>
> (i) Reaching zero emissions for metabolic systems is impossible (open/living systems must breathe),
>
> (ii) The economic process is entropic and therefore a circular economy enabling perpetual economic growth is impossible (it is the biosphere and not the technosphere that recycles matter, while energy cannot be recycled), and
>
> (iii) The existing technosphere has been built on and relied on fossil fuels for over 200 years, and it cannot be completely replaced in 30 years reaching zero emissions while fulfilling all the sustainable development goals (including rapid economic growth in the developing world).
>
> *— M. Giampietro and S.O. Funtowicz [104]*

3.3.2. Storyline narrative

A circular economy focuses on significantly enhancing resource efficiency. This can be achieved through four primary strategies:

1. Minimizing resource use (narrowing the loop) through shared utilization or opting out of certain product use, along with improved manufacturing efficiencies;
2. Prolonging the life and utility of products and their components (slowing the loop) via reuse and repair, thereby reducing the need for new raw materials;
3. Recycling materials (closing the loop) to diminish the quantity of material incinerated or sent to landfills, consequently cutting down the demand for new raw materials;
4. Replacing finite resources with renewable ones (like bioresources) or alternative primary resources that have a lower environmental impact.

In this scenario, we move in the direction of the maximum achievable state of material efficiency as government policy, private innovation and social changes are rapidly driving the transition toward a circular economy. The emphasis here rests heavily on re-X strategies that are implemented in the design phase of products (e.g., repairability and re-manufacturability) and that are actualised by changes in consumer behaviour (e.g reduction, refusal, engagement in the 'sharing economy' and curtailment of the 'throw-away' mindset).

Figure 3.3 presents a simplified visual depiction of the dominant material flow paths in the recovery scenario.

Further, being enabled by the widespread adoption of 'circular design' principles and improvements in information transparency (e.g., waste tracking and digital product passports) the system for the treatment of post-consumer waste can divert a significant amount of their inflows (to, for example, re-use and re-manufacture) with the residual fraction being readily segregated into purer, more efficiently recoverable, material streams.

This scenario envisions a future where government policies are in synergy with private sector innovation and societal changes, driving a wholesale transition towards a circular economy. Unlike the recovery scenario,

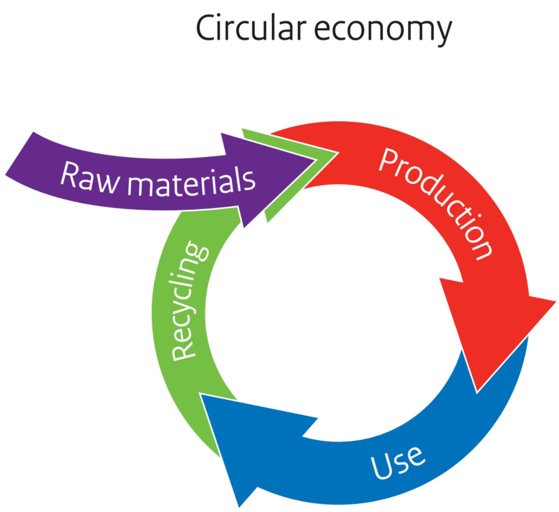

Figure 3.3: The simplified circular economy model in the circularity scenario [9]

where the focus is on the end-of-life recovery of materials, this scenario emphasises minimising waste at all stages, starting from the design phase itself, where both policymakers and designers are moving away from short-lived products towards products designed for longevity.

The emphasis is on re-X strategies that are integrated throughout the entirety of a product's lifecycle. This includes repairability, where products are designed to be easily fixed rather than replaced; and re-manufacturability, where products or their components are designed to be restored to their original state, extending their lifespan and reducing the need for new resources. This scenario calls for a drastic change in consumer behaviour, where reduction in consumption and waste, refusal of non-sustainable options, and active participation in the 'sharing economy' become the norm rather than the exception.

In the circularity scenario, the widespread adoption of 'circular design' principles becomes a cornerstone of production. In a circular design approach, products are designed and produced in a way that considers their entire lifecycle, including eventual disassembly and reuse. New economic models make it costly for producers to generate short-lived products and material waste. Companies are now giving priority to the design of products that are easily repairable, can be disassembled, and reused. The rise of technology has paved the way for predictive maintenance tools. These allow businesses to keep a tab on material conditions through sensors and carry out repairs before a malfunction occurs, a method gaining traction in transport and manufacturing.

Additionally, this scenario envisions an improvement in transparency, with measures such as waste tracking and digital product passports becoming standard. Waste tracking allows for efficient management of waste flows, aiding in effective resource planning, while digital product passports provide information about a product's composition and how it can be properly disassembled, reused, or recycled. Material composition, including raw materials, is transparent to all involved in the value chain, promoting closer collaboration. Producers see the advantage of being open about their product details to aid in repair, repurposing, and recycling activities. This transparency about product components, durability, and reparability increases consumer demand for products that are designed to last and can be reused or recycled.

3.3.3. Scenario needs and impacts

In the proposed scenario, the European Union (EU) embarks on a pivotal transition towards a circular economy. This framework emphasises the retention of product, material, and resource value within the economic matrix for extended durations, simultaneously minimising waste generation. This transition is integral to the EU's strategic goal of cultivating a sustainable, low-carbon, resource-efficient, and globally competitive economy.

The implications of this shift are multifaceted. It presents an avenue for the EU to rejuvenate its economic architecture while providing businesses with a protective shield against challenges such as resource scarcity and price volatility. This revised economic model fosters the emergence of efficient, innovative production and consumption methods, thus offering novel business opportunities. Moreover, the circular economy approach has palpable socio-economic benefits, including diverse job creation and enhanced social integration.

From an environmental perspective, the transition aids in the reduction of the cumulative energy footprint and helps mitigate irreversible ecological damages. This encompasses challenges related to climate shifts, biodiversity conservation, and comprehensive pollution control. Several studies accentuate the overarching benefits of this economic approach, highlighting potential reductions in prevalent carbon dioxide emissions.

The successful implementation of this vision necessitates a collaborative approach involving various stakeholders, encompassing businesses, consumers, and regulatory entities. A robust regulatory framework is indispensable, designed to promote optimal practices and delineate clear progression benchmarks. This comprehensive framework encompasses the entirety of the circular economy's value chain, from production to consumption, extending into realms of repair, remanufacturing, and waste management, culminating in the reintroduction of secondary raw materials into the economic cycle.

Environmental fiscal reforms are crucial for a circular economy transition. Taxes should pivot from labour to resource depletion, promoting a double dividend. The EU can leverage the VAT directive and the European semester process to endorse flexible rates on circular services like repair. It's imperative to abolish harmful subsidies, notably on fossil fuels, which are inherently linear and which Member States have pledged to eliminate. The tax framework should incentivise pioneers challenging the established linear economy. Analysing tax shifts at the national level can determine tax effectiveness and pinpoint instruments that best bolster circularity.

The contribution of member states is paramount. They play a dual role, both in the actualisation of EU directives and in the integration of complementary regional initiatives. The principles of a circular economy possess global applicability, necessitating harmonised strategies within the EU and with external international partners. Such synergised efforts are crucial for the fulfilment of broader international commitments, notably the U.N. 2030 Agenda for Sustainable Development [12]. The ultimate objective remains the establishment of a sustainable future characterised by judicious consumption and production protocols.

3.3.4. Waste stream specific scenario impacts

BATT (Battery waste)

Sources: [24, 86, 7, 8]

In the circularity scenario, battery waste treatment undergoes a massive transformation. The shift towards electric vehicles and renewable energy storage significantly increases the quantity of end-of-life batteries. However, thanks to new regulations, technological advancements, and business models, the majority of battery components are recycled or reused.

- Massive increase in end-of-life batteries due to the shift to electric vehicles and renewable energy storage.
- New regulations incentivise battery manufacturers to design for recycling.
- Battery recycling technologies improve, enabling higher recovery rates of valuable metals.
- Standardised collection systems for battery waste are established, improving the efficiency of the recycling process.
- Service-based business models like leasing ensure manufacturers retain ownership of the batteries, promoting circularity.
- Greater transparency through digital product passports aids in effective battery waste management.
- Battery passport and publicly accessible Information from the new Battery Regulation (SoH, SoC, Predicted lifetime/warranty, etc.) given by the economic operator that places the battery on the market enables high re-use rates.
- Increased repairability/modularity.
- Reduced demand from 'sharing economy' and more 'sustainable' transport choices.
- New emerging technologies more suited for reuse/repair.
- Ambitious targets set by business and public policy.

ELV (End-of-Life Vehicles)

Sources: [86, 142, 30, 31, 32, 33]

For End-of-Life Vehicles (ELVs), the circular economy model affects the way vehicles are designed, used, and discarded. Emphasising extended vehicle life through repair and remanufacturing, this scenario also focuses on the recovery of materials from vehicles at the end of their life.

- Vehicle design shifts towards repairability, upgradability, and recyclability, increasing the lifespan of vehicles.
- Standardised systems for ELV collection are established, ensuring efficient waste management.
- Innovative technologies enable higher recovery rates of metals and other valuable materials from ELVs.
- Service-based models like vehicle leasing and sharing could reduce the total number of vehicles produced.

- Digital product passports provide information about vehicle components, aiding in effective recycling or reuse.
- Focus on managing the use-phase of vehicles.
- Circular strategies take place before material recovery so that material recovery is "delayed".
- Information available to enable these strategies.
- EU vehicles policy has implications for materials in vehicles, such as 'lightweighting' and down-sizing
    - Increase in average occupancy and average vehicle-kilometres per trip.
    - Decrease in average lifetime (in terms of years): As the utilisation factor increases.
- Increase in circular strategies due to an increase in participation from the public and businesses, i.e., target-based incentives with strong regulations and monitoring.

WEEE (Waste Electrical and Electronic Equipment)

Sources: [5, 6, 34, 35, 36, 78]

In the circularity scenario, WEEE becomes a valuable resource instead of a disposal challenge. Thanks to product design changes and the application of advanced recovery technologies, a significant percentage of the materials in WEEE is reclaimed and fed back into the production cycle.

- Electronic products are designed for longevity, repairability, upgradability, and recyclability.
- Advanced technologies enable higher recovery rates of precious metals from WEEE.
- Collection systems for WEEE are improved, ensuring a steady supply of materials to feed the recovery system.
- Digitalisation and data use enhance traceability and efficiency in WEEE management.
- Service-based models for electronics promote the use of products as a service rather than ownership, reducing WEEE generation [89].
- Increased durability and lifespans.
- Increased repairability.
- More sharing and product-service systems, correspond to a reduction in the lifetime (for some equipment).
- More reuse practices (expanded second-hand market).
- Less hoarding.
- Higher formal collection and recycling rate.
- Focus is given more to the production and use phase rather than the EoL (End-of-Life).
- 'Design for circularity' principle: Ecodesign mandates repairability, durability, no obsolescence, modularity, and that continual software upgrades are possible [13, 147].
- Electronically compatible chargers and battery packs can be used by different products.
- The above also means that chargers and batteries are not integrated into the product and that the product is designed to be easily disassembled.
- Strong regulations and monitoring are in place with higher reuse and circular targets, which are set and implemented, and fines are imposed on the member states that fail to achieve the targets.

- Support and development of circular strategies infrastructure (e.g., easy information access for repairability, repair shops, accessibility to spare components on the market, etc.).
- Greater use of connected products, smart technologies, and the IoT. Used to monitor and diagnose product performance in situ which, can extend product and component life.

MIN (Mining Waste)

Sources:

In this scenario, the impact on mining waste is two-fold. Firstly, the need for primary mining is reduced due to lower demand, efficient resource use and high recovery rates of materials. Secondly, mining waste itself is treated as a valuable resource, with advanced technologies being used to extract residual valuable materials.

- A Decrease in primary mining reduces the generation of mining waste.
- Advanced technologies are employed to extract valuable materials from mining waste.
- Policies and regulations incentivise the reuse of mining waste in various applications.
- Digital solutions improve tracking and management of mining waste.
- Collaboration between stakeholders promotes circular practices in the mining industry.

CDW (Construction and Demolition Waste)

Sources: [10]

Construction and Demolition Waste (CDW) is another sector that sees significant improvement in the circularity scenario. This scenario reduces the generation of CDW and promotes the recovery of valuable materials from the waste stream.

- Less demolition and new construction results in a reduction of CDW.
- Buildings are designed for disassembly and reuse, increasing the lifespan of materials and reducing CDW.
- Longer lifetimes for buildings (more renovation and refurbishment) and wind turbines (less repowering, i.e. changing of wind turbines before the end of theoretical lifespan).
- Wind turbine blades are refurbished or repaired and then reused.
- Recycling technologies for CDW improve, allowing higher recovery rates of materials and less 'downcycling'.
- Policies and regulations incentivise the use of recycled materials in construction.
- Standardised systems for CDW collection and separation are improved.
- Digital tools like building information modelling (BIM) improve resource management in construction and renovation.
- Focus on dismantling and selective deconstruction: constructions are taken apart in a way that individual parts can be reused.

SLASH (Slags and Ashes)

Sources: [143, 144, 101, 145, 102]

In the circularity scenario, SLASH are recognised as a potential resource for secondary raw materials. Advances in recovery technologies enable the extraction of valuable metals from SLASH, however, the total volume of CRMs recovered from this material remains low, except in cases of supply constraint. In the circularity scenario, the conditions are very similar to that of the recovery scenario, except that volumes will decrease, due to a general decrease in waste generation.

- Digital solutions enhance the traceability and management of SLASH.
- More functional collection infrastructure.
- Financial support provided to recyclers/operators.
- Introduction of SRM/CRM recovery targets. For example, recovery of P from biomass ash for fertiliser.
- Higher awareness and participation of relevant sectors on SLASH issues and management.
- Strong regulations and monitoring are in place with higher collection and recycling targets.

SLAGS

- Advanced recovery technologies allow for the extraction of valuable metals and minerals from slags.
- New recovery technologies are installed in the metallurgical industry.
- All metals are recovered from the slags, and the slag itself only contains minerals or metals as trace elements.
- Due to the low metal content, the slags are ideal resources for the construction industry.

ASHES

- New recovery technologies are installed in the incineration industry.
- We expect a shift in the volume and quality of the ashes.
- Coal fly ashes are the same as in BAU reducing over time to almost zero.
- Biomass ashes have the same volume and quality.
- Waste incineration ashes: due to the recovery scenario there will be a shift from landfill and incineration towards recycling, reuse, and repair. In countries with a high landfill rate, this will lead to higher volumes of ashes. In countries with a low landfill rate, and a high volume of incineration: this will lead to lower volumes of ashes, and a shift in the composition (with less CRMs in the ashes, due to better pre-sorting)

Table 4.3: Raw materials essential to the renewable energy sector

SUPPLY RISK	MATERIAL	CRM	BATT	H2	WIND	SOLAR (PV)	HEAT PUMPS
5.3	HREE (rest)	●		●			
4.8	Gallium	●				●	
4.4	Niobium	●		●	●		
4.1	Magnesium	●		●			
4.1	REE (magnets)	●		●	●		●
3.8	Boron	●		●	●	●	●
3.5	LREE (rest)	●		●			
3.3	Phosphorus	●	●			●	
2.7	PGM	●		●			
2.6	Strontium	●		●			
2.4	Scandium	●		●			
2.3	Vanadium	●		●			
1.9	Lithium	●	●				
1.8	Geranium	●				●	
1.8	Natural graphite	●	●	●			
1.8	Antimony	●				●	
1.7	Cobalt	●	●	●			
1.6	Arsenic	●				●	
1.4	Silicon metal	●		●	●	●	●
1.3	Baryte	●		●			
1.3	Tantalum	●		●			
1.2	Manganese	●	●	●			●
1.2	Tungsten	●		●			
1.2	Aluminium	●	●	●	●	●	●
1.1	Fluorspar	●	●			●	●
0.9	Tin			●		●	
0.8	Molybdenum			●	●	●	●
0.8	Silver			●		●	●
0.8	Zirconium			●			
0.7	Chromium			●	●		●
0.7	Potash			●			
0.6	Indium					●	
0.5	Nickel	●	●	●	●	●	●
0.5	Iron ore		●	●	●	●	●
0.5	Titanium			●			
0.4	Gold			●			●
0.3	Tellurium					●	
0.3	Limestone			●			

Continued on next page

Table 4.3 — Continued from previous page

SUPPLY RISK	MATERIAL	CRM	BATT	H2	WIND	SOLAR (PV)	HEAT PUMPS
0.3	Selenium					●	
0.3	Silica				●	●	
0.2	Cadmium					●	
0.2	Zinc			●	●	●	●
0.1	Copper	●	●	●	●	●	●
0.1	Aggregates				●		
0.1	Lead				●	●	

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References

Literature referred to in section 13.4 is excluded from the following lists of references, except for those titles cited elsewhere in the report.

Chapter Contents

12.1. Legislation and Government Documents

[1] European Commission. *European Critical Raw Materials Act*. European Commission, 2023. URL: (cit. on pp. vii, 5, 7).

[2] European Commission. *A New Circular Economy Action Plan For a Cleaner and More Competitive Europe*. European Commission, 2020. DOI: 10.2779/05068. URL: (cit. on pp. xvii, 6, 147).

[3] European Commission. *The European Green Deal*. European Commission, 2019. URL: (cit. on pp. 5, 6, 87, 156).

[4] European Commission. *Directive 2002/96/EC Of The European Parliament and of The Council of 27 January 2003 on waste electrical and electronic equipment (WEEE)*. European Commission, 2003. URL: (cit. on p. 5).

[5] European Commission. *Directive 2012/19/EU of the European Parliament and of the council of 4 July 2012 on waste electrical and electronic equipment (WEEE) text with EEA relevance*. European Commission, 2012. URL: (cit. on pp. 5, 29, 36, 45, 134).

[6] European Commission. *Directive 2012/19/EU of the European Parliament and of the council of 4 July 2012 on waste electrical and electronic equipment (WEEE) (recast)*. European Commission, 2012. URL: (cit. on pp. 5, 29, 36, 45, 134).

[7] European Commission. *Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries, amending Directive 2008/98/EC and Regulation (EU) 2019/1020 and repealing Directive 2006/66/EC*. European Commission, 2023. URL: (cit. on pp. 5, 28, 35, 44).

[8] European Commission. *Proposal for a Regulation of the European Parliament and of the Council Concerning Batteries and Waste Batteries, Repealing Directive 2006/66/EC and Amending Regulation (EU) No 2019/1020*. European Commission, 2020. URL: (cit. on pp. 6, 7, 28, 35, 44).

[9] Government of the Netherlands. *A Circular Economy in the Netherlands by 2050*. 2016. URL: (cit. on pp. 26, 33, 42).

[10] European Commission. *Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (Text with EEA relevance)*. European Commission, 2008. URL: (cit. on pp. 29, 30, 37, 46, 133).

[11] Government of the Netherlands. *National Circular Economy Programme 2023-2030*. 2023. URL: (cit. on pp. 40, 120).

[12] United Nations General Assembly. *Transforming our world: the 2030 Agenda for Sustainable Development*. 2015. URL: (cit. on p. 43).

[13] European Commission. *Directive 2014/53/EU of the European Parliament and of the Council of 16 April 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment and repealing Directive 1999/5/EC*. European Commission, 2023. URL: (cit. on p. 45).

[14] European Parliament and Council of the European Union. *Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023 amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC as regards the promotion of energy from renewable sources, and repealing Council Directive (EU) 2015/652*. Directive. European Parliament and Council of the European Union, Oct. 31, 2023. URL: (cit. on p. 82).

[15] European Commission. *Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions: REPowerEU Plan*. Communication. European Commission, 2022. URL: (cit. on pp. 82, 87).

[16] European Commission. *Fit for 55 - The EU's plan for a green transition - Consilium*. 2021. URL: (cit. on p. 87).

[17] European Parliament and Council of the European Union. *Draft Directive of the European Parliament and of the Council amending Directive (EU) 2018/2001, Regulation (EU) 2018/1999 and Directive 98/70/EC as regards the promotion of energy from renewable sources, and repealing Council Directive (EU) 2015/652 (first reading) - Interinstitutional File: 2021/0218(COD)*. Interinstitutional File. European Parliament and Council of the European Union, Oct. 6, 2023. URL: (cit. on pp. 88, 89).

[18] European Commission. *Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions on a Monitoring Framework for the Circular Economy*. COM (2018) 29 Final. 2018. URL: (visited on 11/02/2018) (cit. on pp. 120, 122, 129).

[19] European Commission. *Commission Staff Working Document: Measuring Progress towards Circular Economy in the European Union – Key Indicators for a Revised Monitoring Framework Accompanying the Document Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions on a Revised Monitoring Framework for the Circular Economy*. Staff Working Document. SWD/2023/306 final. 2023. URL: (cit. on pp. 122, 123, 129).

[20] Directorate-General for Justice European Commission and Consumers. *Proposal for a Directive of the European Parliament and of the Council on Common Rules Promoting the Repair of Goods and Amending Regulation (EU) 2017/2394, Directives (EU) 2019/771 and (EU) 2020/1828*. Directive Proposal. 2023. URL: (cit. on p. 147).

[21] Ministère de la transition écologique. *Décret n°2020-1757 du 29 décembre 2020 relatif à l'indice de réparabilité des équipements électriques et électroniques [Decree No. 2020-1757 of 29 December 2020 on the repairability index of electrical and electronic equipment]*. 2020 (cit. on p. 148).

12.2. REPORTS

[22] European Commission et al. *Study on the critical raw materials for the EU 2023 —– Final report*. 2023. DOI: 10.2873/725585. URL: (cit. on p. vii).

[23] Lena Börjeson et al. *Towards a user's guide to scenarios - a report on scenario types and scenario techniques*. 2005. URL: (cit. on pp. xvi, 3, 10, 12).

[24] Vivienne Halleux. *New EU regulatory framework for batteries: Setting sustainability requirements*. 2021. URL: (cit. on pp. xvii, 28, 35, 44).

[25] International Energy Agency (IEA). *Critical Minerals Market Review 2023*. 2023. URL: (cit. on pp. xviii, 86, 92).

[26] Lucia Mancini, Camillo De Camillis, and David Pennington. *Security of supply and scarcity of raw materials: Towards a methodological framework for sustainability assessment*. 2013. DOI: 10.2788/94926. URL: (cit. on p. xviii).

[27] Marcin Szczepański. *Resilience of global supply chains: Challenges and solutions*. 2021. URL: (cit. on p. 8).

[28] International Energy Agency (IEA). *World Energy Outlook 2022*. 2022. URL: (cit. on p. 25).

[29] Hoesung Lee, Katherine Calvin, Dipak Dasgupta, et al. *Climate Change 2023 Synthesis Report Summary for Policymakers*. 2024. URL: (cit. on p. 26).

[30] A. Løvik, C. Marmy, M. Ljunggren, et al. *Material composition trends in vehicles: critical raw materials and other relevant metals*. 2021. DOI: 10.2760/351825. URL: (cit. on pp. 28, 35, 44).

[31] N. Tazi et al. *Initial analysis of selected measures to improve the circularity of critical raw materials and other materials in vehicles*. 2023. DOI: 10.2760/207541. URL: (cit. on pp. 28, 35, 44).

[32] L. Ljunggren, A. Løvik, D. Kushnir, et al. *RMIS – Raw materials in vehicles*. 2022. URL: (cit. on pp. 28, 35, 44).

[33] International Energy Agency (IEA). *Global EV Outlook 2023*. 2022. URL: (cit. on pp. 28, 35, 44).

[34] K. Parajuly, R. Kuehr, A. K. Awasthi, et al. *Future E-waste Scenarios*. StEP Initiative, UNU ViE-SCYCLE, UNEP IETC, 2019. URL: (cit. on pp. 29, 36, 45).

[35] United Nations Institute for Training United Nations Environment Programme and Research. *Electronic and Electrical Waste Outlook in West Asia*. 2023. URL: (cit. on pp. 29, 36, 45).

[36] Vanessa Forti, Cornelis P Balde, Ruediger Kuehr, et al. *The Global E-waste Monitor 2020: Quantities, flows and the circular economy potential*. 2020. URL: (cit. on pp. 29, 36, 45).

[37] Jaco Huisman et al. *Countering WEEE Illegal Trade (CWIT) Summary Report, Market Assessment, Legal Analysis, Crime Analysis and Recommendations Roadmap*. 2015. DOI: 10.13140/rg.2.1.4864.2328. URL: (cit. on p. 37).

[38] Ellen MacArthur. *Towards a circular economy: Business rationale for an accelerated transition*. 2015. URL: (cit. on pp. 40, 41).

[39] B. Oberle et al. *Global Resources Outlook 2019: Natural Resources for the Future We Want*. Tech. rep. United Nations Environment Programme, 2019. URL: (cit. on p. 65).

[40] Yvan Guillemette and David Turner. *The long game: Fiscal outlooks to 2060 underline need for structural reform*. 2021. DOI: 10.1787/a112307e-en. URL: (cit. on pp. 76, 77).

[41] Robert J. Barro. *Determinants of Economic Growth in a Panel of Countries*. CEMA Working Papers. China Economics, Management Academy, Central University of Finance, and Economics, 2003. URL: (cit. on p. 76).

[42] Jean Chateau et al. *Long-term economic growth and environmental pressure: reference scenarios for future global projections*. 2012 (cit. on p. 76).

[43] Stefano Carrara et al. *Supply chain analysis and material demand forecast in strategic technologies and sectors in the EU – A foresight study*. Publications Office of the European Union, 2023. DOI: 10.2760/334074. URL: (cit. on pp. 82, 86).

[44] Keramidas K et al. *Global Energy and Climate Outlook 2022: Energy trade in a decarbonised world*. Policy assessment. 2022. DOI: 10.2760/863694(online) (cit. on p. 82).

[45] International Energy Agency (IEA). *Renewables 2022*. 2022. URL: (cit. on pp. 82, 89).

[46] International Energy Agency (IEA). *Energy Technology Perspectives 2023*. 2023. DOI: 10.1787/7c6b23db-en. URL: (cit. on p. 86).

[47] European Commission. *Commission Staff Working Document Implementing the REPowerEU Action Plan: Investment Needs, Hydrogen Accelerator and Achieving the Bio-methane Targets Accompanying the document Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions REPowerEU Plan COM(2022) 230 final*. Commission Staff Working Document. European Commission, May 18, 2022. URL: (cit. on p. 89).

[48] European Commission et al. *EU reference scenario 2020 — Energy, transport and GHG emissions — Trends to 2050*. 2021. DOI: doi/10.2833/35750. URL: (cit. on p. 90).

[49] Mantzos L et al. *The POTEnCIA Central scenario: an EU energy outlook to 2050*. 2021. DOI: 10.2760/32835. URL: (cit. on p. 90).

[50] R. Pardo and J.P. Schweitzer. *A Long-term Strategy for a European Circular Economy — Setting the Course for Success*. Policy Paper. Produced for the Think2030 project. Think2030, 2018 (cit. on p. 120).

[51] OECD. *Material Resources, Productivity and the Environment*. 2015. DOI: https://doi.org/https://doi.org/10.1787/9789264190504-en. URL: (cit. on p. 120).

[52] Aldert Hanemaaijer et al. *Possible Objectives for a Circular Economy: Summary of the Dutch policy brief 'Mogelijke doelen voor een Circulaire Economie'*. PBL Netherlands Environmental Assessment Agency, 2021. URL: (cit. on pp. 120, 128).

[53] M. Cordella, F. Alfieri, and J. Sanfelix Forner. *Analysis and development of a scoring system for repair and upgrade of products*. Jrc114337. Publications Office of the European Union, 2019. DOI: 10.2760/725068 (cit. on pp. 145–149).

[54] M. Moeslinger et al. *Towards an Effective Right to Repair for Electronics*. JRC Technical Reports. Publications Office of the European Union, 2022. DOI: 10.2760/42722. URL: (cit. on pp. 145, 147, 149).

[55] Christoforos Spiliotopoulos et al. *Product reparability scoring system: Specific application to smartphones and slate tablets*. 2022 (cit. on p. 147).

[56] David Parker et al. *Remanufacturing market study*. 2015. URL: (cit. on pp. 156, 157).

[57] Nabil Nasr et al. *Re-defining Value — The Manufacturing Revolution. Remanufacturing, Refurbishment, Repair and Direct Reuse in the Circular Economy*. Tech. rep. United Nations Environment Programme, 2018. URL: (cit. on p. 156).

[58] Robert Vaughan and Raphael Daverio. *PWC report for DG-Grow: Analytical paper on the economic scale and growth of the collaborative economy*. 2016. URL: (cit. on pp. 164, 165).

[59] European Commission et al. *Study to monitor the economic development of the collaborative economy at sector level in the 28 EU Member States — Final report*. 2018. DOI: doi/10.2873/83555. URL: (cit. on p. 164).

[60] European Commission et al. *Environmental potential of the collaborative economy — Final report and annexes*. 2018. DOI: doi/10.2779/518554. URL: (cit. on pp. 164, 165).

[61] Alberta Andreotti et al. *Participation in the Sharing Economy : Report from the EU H2020 Research Project Ps2Share: Participation, Privacy, and Power in the Sharing Economy*. 2017. DOI: 10.5167/uzh-149823. URL: (cit. on p. 164).

[62] Intergovernmental Panel on Climate Change (IPCC). *Climate Change and Land*. 2022. DOI: 10.1017/9781009157988. URL: (cit. on p. 198).

12.3. DATASETS

[63] Eurostat and European Commission. *Population projections - Population and demography - Eurostat*. 2023. URL: (cit. on pp. 66, 74).

[64] Office for National Statistics — ONS. *2020-based interim national population projections: year ending June 2022 estimated international migration variant*. (Accessed on 10/30/2023). 2023. URL: (cit. on pp. 66, 68, 74).

[65] OECD. *Long-term baseline projections, No. 109 (Edition 2021)*. Accessed on 01 November 2023. 2021. DOI: 10.1787/cbdb49e6-en. URL: (cit. on pp. 74, 76).

[66] *Databank | The World Bank*. The World Bank. 2011. URL: (cit. on p. 76).

[67] European Commission, Joint Research Centre. *Forecast of Rare Earths Supply and Demand - Wind Energy and E-Mobility*. Updated 2020-12-17, Accessed 2023-11-13. 2020. URL: (cit. on pp. 86, 92).

[68] International Energy Agency (IEA). *World Energy Outlook 2022*. International Energy Agency (IEA). 2022. URL: (cit. on p. 90).

[69] Eurostat. *Database - Eurostat*. 2023. URL: (cit. on p. 129).

12.4. Books

[70] Julian Lincoln Simon. *The ultimate resource 2*. Princeton University Press, 1998 (cit. on p. 25).

[71] Julian Lincoln Simon. *The ultimate resource*. Princeton University Press, 1983 (cit. on p. 25).

[72] Paul R Ehrlich, Anne H Ehrlich, and John P Holdren. *Ecoscience: population, resources, environment*. 3rd ed. W.H. Freeman, 1978 (cit. on p. 25).

[73] Paul R Ehrlich. *The Population Bomb*. Random House, 1986 (cit. on p. 25).

[74] Robert J. Barro and Xavier Sala-i-Martin. *Economic growth*. 2nd ed. The MIT Press, 2004. ISBN: 0-262-02553-1. URL: (cit. on p. 76).

[75] Walter Stahel. *The Circular Economy - a user's guide*. 2019. ISBN: 978-0-367-20014-5 (cit. on p. 120).

12.5. BOOK CHAPTERS

[76] Saskia Sardesai, Markus Stute, and Josef Kamphues. "A Methodology for Future Scenario Planning". In: *Next Generation Supply Chains: A Roadmap for Research and Innovation*. Ed. by Rosanna Fornasiero et al. Springer International Publishing, 2021. ISBN: 978-3-030-63505-3. DOI: 10.1007/978-3-030-63505-3_2. URL: (cit. on pp. xvi, 3).

[77] Manel Sansa, Ahmed Badreddine, and Taieb Ben Romdhane. "Sustainable design based on LCA and operations management methods: SWOT, PESTEL, and 7S". In: *Methods in sustainability science*. Elsevier, 2021. DOI: 10.1016/b978-0-12-823987-2.00019-2. URL: (cit. on p. 16).

[78] Isabel Narbón-Perpiñá and Diego Prior. "Management of Waste Electrical and Electronic Equipment in European Union Countries: A Comparison". In: *E-waste Recycling and Management: Present Scenarios and Environmental Issues*. Ed. by Anish Khan, Inamuddin, and Abdullah M. Asiri. Springer International Publishing, 2020. ISBN: 978-3-030-14184-4. DOI: 10.1007/978-3-030-14184-4_7. URL: (cit. on pp. 29, 36, 45).

[79] Warren E. Walker, Robert J. Lempert, and Jan H. Kwakkel. "Deep Uncertainty". In: *Encyclopedia of Operations Research and Management Science*. Ed. by Saul I. Gass and Michael C. Fu. Springer US, 2013. ISBN: 978-1-4419-1153-7. DOI: 10.1007/978-1-4419-1153-7_1140. URL: (cit. on p. 198).

12.6. Articles

[80] T. E. Graedel, E. M. Harper, N. T. Nassar, et al. "On the materials basis of modern society". In: *Proceedings of the National Academy of Sciences* (2015). DOI: 10.1073/pnas.1312752110. URL: (cit. on p. vii).

[81] Karl H Dreborg. "Essence of backcasting". In: *Futures* (1996). DOI: 10.1016/s0016-3287(96)00044-4. URL: (cit. on pp. xvi, 3).

[82] Lena Börjeson et al. "Scenario types and techniques: Towards a user's guide". In: *Futures* (2006). DOI: 10.1016/j.futures.2005.12.002. URL: (cit. on pp. xvi, 3, 10, 12).

[83] Muhammad Amer, Tugrul U. Daim, and Antonie Jetter. "A review of scenario planning". In: *Futures* (2013). DOI: 10.1016/j.futures.2012.10.003. URL: (cit. on pp. xvi, 3, 5, 10).

[84] Elisabeth A Lloyd and Vanessa J Schweizer. "Objectivity and a comparison of methodological scenario approaches for climate change research". In: *Synthese* (2014). DOI: 10.1007/s11229-013-0353-6. URL: (cit. on p. xvi).

[85] Fulvio Ardente and Fabrice Mathieux. "Environmental assessment of the durability of energy-using products: method and application". In: *Journal of Cleaner Production* (2014). DOI: 10.1016/j.jclepro.2014.03.049. URL: (cit. on p. xvi).

[86] Harald Helander and Maria Ljunggren. "Battery as a service: Analysing multiple reuse and recycling loops". In: *Resources, Conservation and Recycling* (2023). DOI: 10.1016/j.resconrec.2023.107091. URL: (cit. on pp. xvii, 28, 35, 44).

[87] Julian Kirchherr, Denise Reike, and Marko Hekkert. "Conceptualizing the circular economy: An analysis of 114 definitions". In: *Resources, Conservation and Recycling* (2017). DOI: 10.1016/j.resconrec.2017.09.005. URL: (cit. on pp. xvii, 120).

[88] Teresa Domenech and Bettina Bahn-Walkowiak. "Transition towards a resource efficient circular economy in Europe: policy lessons from the EU and the member states". In: *Ecological Economics* (2019) (cit. on p. xvii).

[89] Martin Geissdoerfer, Marina P.P. Pieroni, Daniela C.A. Pigosso, et al. "Circular business models: A review". In: *Journal of Cleaner Production* (2020). DOI: 10.1016/j.jclepro.2020.123741. URL: (cit. on pp. xx, 45).

[90] Peter Bishop, Andy Hines, and Terry Collins. "The current state of scenario development: An overview of techniques". In: *Foresight - The journal of future studies* (2007). DOI: 10.1108/14636680710727516 (cit. on pp. 3, 10).

[91] Kathya Cordova-Pozo and Etiënne A.J.A. Rouwette. "Types of scenario planning and their effectiveness: A review of reviews". In: *Futures* (2023). DOI: 10.1016/j.futures.2023.103153 (cit. on pp. 3, 10).

[92] Jim Skea, Renée van Diemen, Joana Portugal-Pereira, et al. "Outlooks, explorations and normative scenarios: Approaches to global energy futures compared". In: *Technological Forecasting and Social Change* (2021) (cit. on pp. 3, 10, 196–198).

[93] Philip W.F van Notten, Jan Rotmans, Marjolein B.A van Asselt, et al. "An updated scenario typology". In: *Futures* (2003). DOI: 10.1016/s0016-3287(02)00090-3. URL: (cit. on p. 3).

[94] Matthew Spaniol and Nicholas Rowland. "The scenario planning paradox". In: *Futures* (2017). DOI: 10.1016/j.futures.2017.09.006. URL: (cit. on p. 5).

[95] Herman Kahn and Anthony J. Wiener. "The Next Thirty-Three Years: A Framework for Speculation". In: *Daedalus* (1967). URL: (cit. on p. 5).

[96] Bernhard Steubing, Christopher Mutel, Florian Suter, et al. "Streamlining scenario analysis and optimization of key choices in value chains using a modular LCA approach". In: *The International Journal of Life Cycle Assessment* (2016) (cit. on p. 10).

[97] Chia-Chien Hsu and Brian A. Sandford. "The Delphi Technique: Making Sense of Consensus". In: *Practical Assessment, Research, and Evaluation* (2019). DOI: 10.7275/pdz9-th90. URL: (cit. on p. 10).

[98] Konstantinos Kokkinos, Vayos Karayannis, Nicholas Samaras, et al. "Multi-scenario analysis on hydrogen production development using PESTEL and FCM models". In: *Journal of Cleaner Production* (2023). DOI: 10.1016/j.jclepro.2023.138251. URL: ↗ (cit. on p. 16).

[99] Paul R Ehrlich and Anne H Ehrlich. "The population bomb revisited". In: *The electronic journal of sustainable development* (2009). URL: ↗ (cit. on p. 25).

[100] Julia Moschen-Schimek, Thomas Kasper, and Marion Huber-Humer. "Critical review of the recovery rates of construction and demolition waste in the European Union — An analysis of influencing factors in selected EU countries". In: *Waste Management* (2023). DOI: 10.1016/j.wasman.2023.05.020. URL: ↗ (cit. on p. 30).

[101] Valentin Vogl, Max Åhman, and Lars J. Nilsson. "The making of green steel in the EU: a policy evaluation for the early commercialization phase". In: *Climate Policy* (2020). DOI: 10.1080/14693062.2020.1803040. URL: ↗ (cit. on pp. 30, 38, 47).

[102] C. Asquer et al. "Biomass ash characterisation for reuse as additive in composting process". In: *Biomass and Bioenergy* (2019). DOI: 10.1016/j.biombioe.2019.03.001. URL: ↗ (cit. on pp. 30, 38, 47).

[103] Robert U Ayres. "Metals recycling: economic and environmental implications". In: *Resources, Conservation and Recycling* (1997). DOI: 10.1016/s0921-3449(97)00033-5. URL: ↗ (cit. on p. 32).

[104] Mario Giampietro and Silvio O. Funtowicz. "From elite folk science to the policy legend of the circular economy". In: *Environmental Science & Policy* (2020). DOI: 10.1016/j.envsci.2020.04.012. URL: ↗ (cit. on p. 41).

[105] Patrizio Vanella, Philipp Deschermeier, and Christina B. Wilke. "An Overview of Population Projections—Methodological Concepts, International Data Availability, and Use Cases". In: *Forecasting* (2020). DOI: 10.3390/forecast2030019. URL: ↗ (cit. on pp. 66, 69).

[106] Detlef P. van Vuuren et al. "The Shared Socio-economic Pathways: Trajectories for human development and global environmental change". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2016.10.009. URL: ↗ (cit. on pp. 66, 77).

[107] Samir KC and Wolfgang Lutz. "The human core of the shared socioeconomic pathways: Population scenarios by age, sex and level of education for all countries to 2100". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2014.06.004 (cit. on pp. 66, 77).

[108] Romain Duval and Christine de la Maisonneuve. "Long-run GDP growth framework and scenarios for the world economy". In: *Journal of Policy Modeling* (2010). Also published as OECD Economics Department Working Papers 663 (cit. on p. 76).

[109] N. Gregory Mankiw, David Romer, and David N. Weil. "A contribution to the empirics of economic growth". In: *Quarterly Journal of Economics* (1992) (cit. on p. 76).

[110] Lionel Fontagné, Erica Perego, and Gianluca Santoni. "MaGE 3.1: Long-term macroeconomic projections of the World economy". In: *International Economics* (2022). URL: ↗ (cit. on p. 76).

[111] Leiwen Jiang and Brian C. O'Neill. "Global urbanization projections for the Shared Socioeconomic Pathways". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2015.03.008 (cit. on p. 77).

[112] Marian Leimbach et al. "Future growth patterns of world regions – A GDP scenario approach". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2015.02.005 (cit. on p. 77).

[113] Jesús Crespo Cuaresma. "Income projections for climate change research: A framework based on human capital dynamics". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2015.02.012 (cit. on p. 77).

[114] Rob Dellink et al. "Long-term economic growth projections in the Shared Socioeconomic Pathways". In: *Global Environmental Change* (2017). DOI: 10.1016/j.gloenvcha.2015.06.004 (cit. on p. 77).

[115] Markus Reuter and Antoinette van Schaik. "Opportunities and limits of recycling: A dynamic-model-based analysis". In: *MRS Bulletin* (2012). DOI: 10.1557/mrs.2012.57. URL: ↗ (cit. on p. 112).

[116] Piero Morseletto. "Targets for a circular economy". In: *Resources, Conservation and Recycling* (2020). DOI: 10.1016/j.resconrec.2019.104553. URL: ↗ (cit. on pp. 120, 140).

[117] Alfonso Marino and Paolo Pariso. "Comparing European countries' performances in the transition towards the Circular Economy". In: *Science of The Total Environment* (2020). DOI: 10.1016/j.scitotenv.2020.138142. URL: ↗ (cit. on p. 120).

[118] Martin Calisto Friant, Walter J.V. Vermeulen, and Roberta Salomone. “Analysing European Union circular economy policies: words versus actions”. In: *Sustainable Production and Consumption* (2021). DOI: https://doi.org/10.1016/j.spc.2020.11.001. URL: (cit. on p. 120).

[119] Denise Reike, Walter J.V. Vermeulen, and Sjors Witjes. “The circular economy: New or Refurbished as CE 3.0? — Exploring Controversies in the Conceptualization of the Circular Economy through a Focus on History and Resource Value Retention Options”. In: *Resources, Conservation and Recycling* (2018). Sustainable Resource Management and the Circular Economy. DOI: https://doi.org/10.1016/j.resconrec.2017.08.027. URL: (cit. on pp. 120, 122, 139, 156).

[120] Walter Vermeulen, Denise Reike, and Sjors Witjes. “Circular Economy 3.0 - Solving confusion around new conceptions of circularity by synthesising and re-organising the 3R’s concept into a 10R hierarchy”. In: *Renewable Matter* (2019). URL: (cit. on p. 139).

[121] Eléonore Maitre-Ekern and Carl Dalhammar. “Towards a hierarchy of consumption behaviour in the circular economy”. In: *Maastricht Journal of European and Comparative Law* (2019). DOI: 10.1177/1023263X19840943. URL: (cit. on p. 140).

[122] Mauro Cordella, Felice Alfieri, and Javier Sanfelix. “Reducing the carbon footprint of ICT products through material efficiency strategies: A life cycle analysis of smartphones”. In: *Journal of Industrial Ecology* (2021). DOI: https://doi.org/10.1111/jiec.13119. URL: (cit. on pp. 145, 149).

[123] H. Wieser and N. Tröger. “Exploring the inner loops of the circular economy: Replacement, repair, and reuse of mobile phones in Austria”. In: *Journal of Cleaner Production* (2018). DOI: 10.1016/j.jclepro.2017.11.106. URL: (cit. on pp. 145, 149).

[124] S. Svensson et al. “The Emerging ‘Right to Repair’ legislation in the EU and the U.S.” In: *Going Green Care Innovation* (2018) (cit. on pp. 145, 146).

[125] Ricardo J Hernandez, Constanza Miranda, and Julian Goñi. “Empowering sustainable consumption by giving back to consumers the ‘right to repair’”. In: *Sustainability* (2020) (cit. on pp. 145, 146).

[126] Laura Ruiz-Pastor and Jaime A. Mesa. “Proposing an integrated indicator to measure product repairability”. In: *Journal of Cleaner Production* (2023). DOI: https://doi.org/10.1016/j.jclepro.2023.136434. URL: (cit. on p. 147).

[127] Mário Barros and Eric Dimla. “Smartphone repairability indexes in practice: Linking repair scores to industrial design features”. In: *Journal of Industrial Ecology* (2023). DOI: 10.1111/jiec.13398 (cit. on p. 147).

[128] Sagar Dangal, Jeremy Faludi, and Ruud Balkenende. “Design Aspects in Repairability Scoring Systems: Comparing Their Objectivity and Completeness”. In: *Sustainability* (2022). DOI: 10.3390/su14148634. URL: (cit. on p. 147).

[129] Stephanie Boulos et al. “The durability of products: standard assessment for the circular economy under the Eco-Innovation Action Plan”. In: *Report for European Commission, DG Environment* (2015) (cit. on p. 148).

[130] Tamar Makov and David Font Vivanco. “Does the circular economy grow the pie? The case of rebound effects from smartphone reuse”. In: *Frontiers in Energy Research* (2018). DOI: 10.3389/fenrg.2018.00039. URL: (cit. on p. 149).

[131] Siru Sihvonen and Tuomas Ritola. “Conceptualizing ReX for Aggregating End-of-life Strategies in Product Development”. In: *Procedia CIRP* (2015). DOI: 10.1016/j.procir.2015.01.026. URL: (cit. on p. 156).

[132] Monsuru O. Ramoni and Hong-chao Zhang. “An entropy-based metric for product remanufacturability”. In: *Journal of Remanufacturing* (2012). DOI: 10.1186/2210-4690-2-2. URL: (cit. on p. 156).

[133] Subrata Mitra. “Revenue management for remanufactured products”. In: *Omega* (2007). DOI: https://doi.org/10.1016/j.omega.2005.10.003. URL: (cit. on pp. 156, 157).

[134] Deepak Singhal, Sushanta Tripathy, and Sarat Kumar Jena. “Remanufacturing for the circular economy: Study and evaluation of critical factors”. In: *Resources, Conservation and Recycling* (2020). DOI: https://doi.org/10.1016/j.resconrec.2020.104681. URL: (cit. on pp. 157, 158).

[135] Jiali Zhao et al. “An energy and time prediction model for remanufacturing process using graphical evaluation and review technique (GERT) with multivariant uncertainties”. In: *Environmental Science and Pollution Research* (2021). DOI: 10.1007/s11356-021-13438-z (cit. on p. 158).

[136] Amal Dabbous and Abbas Tarhini. “Does sharing economy promote sustainable economic development and energy efficiency? Evidence from OECD countries”. In: *Journal of Innovation & Knowledge* (2021). DOI: 10.1016/j.jik.2020.11.001. URL: (cit. on pp. 164, 165).

[137] Xiaoxi Zhu and Kai Liu. “A systematic review and future directions of the sharing economy: business models, operational insights and environment-based utilities”. In: *Journal of Cleaner Production* (2021). DOI: https://doi.org/10.1016/j.jclepro.2020.125209. URL: (cit. on pp. 164, 165).

[138] Andrea Geissinger et al. “How sustainable is the sharing economy? On the sustainability connotations of sharing economy platforms”. In: *Journal of Cleaner Production* (2019). DOI: 10.1016/j.jclepro.2018.09.196. URL: (cit. on p. 165).

[139] Mohammed Mahmoud et al. “A formal framework for scenario development in support of environmental decision-making”. In: *Environmental Modelling & Software* (2009). DOI: 10.1016/j.envsoft.2008.11.010. URL: (cit. on p. 197).

[140] Eva Wollenberg, David Edmunds, and Louise Buck. “Using scenarios to make decisions about the future: anticipatory learning for the adaptive co-management of community forests”. In: *Landscape and Urban Planning* (2000). DOI: 10.1016/s0169-2046(99)00071-7. URL: (cit. on p. 197).

12.7. ONLINE SOURCES

[141] European Commission. *Questions and Answers: The European Critical Raw Materials Act*. European Commission. 2023. URL: (cit. on p. vii).

[142] European Commission. *End-of-life vehicles Regulation*. European Commission. 2023. URL: (cit. on pp. 5, 28, 35, 44).

[143] European Steel Association (Eurofer). *European Steel in Figures*. Accessed: 15 November 2023. 2023. URL: (cit. on pp. 30, 38, 47).

[144] European Aluminium. *Aluminium Industry*. Accessed: 15 November 2023. 2023. URL: (cit. on pp. 30, 38, 47).

[145] Visual Capitalist. *Visualizing 50 Years of Global Steel Production*. Accessed: 15 November 2023. 2023. URL: (cit. on pp. 30, 38, 47).

[146] CBS - Statistics Netherlands. *Circularity of the Dutch economy has barely increased*. 2023. URL: (cit. on p. 40).

[147] European Parliament. *Long-awaited common charger for mobile devices will be a reality in 2024*. European Commission. 2023. URL: (cit. on p. 45).

[148] TÜV SÜD. *Repairability Testing | TÜV SÜD*. 2023. URL: (cit. on p. 148).

[149] Center for Climate Adaptation Science and Solutions at the University of Arizona. *Glossary - Scenario Planning for the Institute of the Environment*. 2023. URL: (cit. on pp. 196–198).

12.8. MISCELLANEOUS

[150] United Nations Economic Commission for Europe (UNECE). *United Nations Framework Classification for Resources (UNFC)*. United Nations Economic Commission for Europe (UNECE), 2023. URL: (cit. on pp. xvi, 3).

[151] FutuRaM consortium European Commission. *Grant Agreement — Project 101058522 — FutuRaM*. 2023. DOI: 10.3030/101058522. URL: (cit. on pp. xxv, xxvii, xxxi, 222).

[152] FutuRaM consortium. *Consortium Agreement — Project 101058522 — FutuRaM*. 2022 (cit. on pp. xxv, 3).

[153] Kees Baldé James Horne. *Project Management Plan — Project 101058522 — FutuRaM*. 2023 (cit. on pp. xxv, 3, 222).

[154] Kirsten Remmen et al. *FutuRaM conceptual and methodological framework (M6 report) — Project 101058522 — FutuRaM*. 2023 (cit. on p. xxv).

[155] Damian Carrington. *Interview with Paul Ehrlich: 'Collapse of civilisation is a near certainty within decades'*. 2018. URL: (cit. on p. 25).

[156] Aurubis. *ReShape The Future Magazine — Aurubis annual report 2018/19*. 2019. URL: (cit. on p. 32).

[157] Randall Munroe. *xkcd: Sustainable*. 2012. URL: (cit. on p. 57).

[158] OECD. *Long-term growth scenarios*. OECD, 2012 (cit. on p. 76).

[159] International Energy Agency (IEA). *Is the European Union on track to meet its REPowerEU goals?* International Energy Agency (IEA), 2022. URL: (cit. on p. 89).

[160] International Energy Agency (IEA). *Global Energy and Climate Model*. International Energy Agency (IEA), 2023. URL: (cit. on p. 90).

[161] European Commission. *Product Bureau | Circular Economy: Environmental and Waste Management*. 2023. URL: (cit. on p. 147).

Appendices

CHAPTER CONTENTS

13.1. TERMINOLOGY

The following is a suggested terminology for use in our discussions and reports related to scenarios.

This glossary is modelled on that used by [92]. Some additional definitions were sourced from [149].

Table 13.1: Terminology

TERM	DEFINITION	LEVEL/CONTEXT	ALSO CALLED	SOURCE
Normative scenario	Goal-oriented scenario: identify decisions and investments that must be made to achieve desired future outcomes. Example: Constraining cumulative emissions	Scenario type	Backcasting	[92]
Exploratory scenario	Exploration of plausible alternative developments to test whether decision-making is robust against different outcomes. Generally, involving a qualitative storyline about a possible future, complemented by quantitative analysis. Example: Socio-economic scenarios	Scenario type	Plausible scenarios	[92]
Outlook	To provide a most likely estimate of future trends as a guide for decision-making	Scenario type	Forecast, projection	[92]
Scenario characteristics	A combination of the vague attributes that make up the qualitative storyline for a scenario. For example, in WEC (2019) the scenario titled Modern Jazz is described as: "A market-led, digitally disrupted world with faster-paced and more uneven economic growth. Recent signals suggest that this entrepreneurial future might accelerate clean energy access on both global and local scales, whilst presenting new systems integration, cyber security and data privacy challenges"	Scenario description	Qualitative storyline descriptors	[92]
Scenario scale	Description of the spatial extent or temporal extent of a scenario. For us, mostly EU toward 2050.	Scenario component		[149]
Scenario dimensions	Uncertainties around which scenarios are constructed, represented as axes in some methods. In our case they might end up being, level of circularity, free-trade/autarky, progress in energy transition	Scenario component		[149]
Scenario literature	Journal articles, grey literature, etc., from which data is sourced that can be used to justify decisions in scenario development	Scenario component		[92]
Scenario logics	Methods for structuring the relationships between different drivers and assumptions in scenarios	Scenario component		[149]
Time horizon	End date of the scenario's forecast	Scenario attribute		[92]
Snapshot	The position of scenario/s at a particular point of time	Scenario attribute		[92]
Storyline and simulation	Combination of qualitative narrative development and quantitative modelling	Scenario component		[139, 140], in [149]

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TERM	DEFINITION	LEVEL/CONTEXT	ALSO CALLED	SOURCE
Marker scenario	Generally, a widely accepted scenario which can be used a guide or to provide background information. E.g., SSP1-5, and the GEC models from the IEA. If applicable, these can be extended upon or combined to help build our models.	Scenario description	Basis scenario	[92]
SSP	Shared Social Pathways. They "describe plausible major global developments that together would lead in the future to different challenges for mitigation and adaptation to climate change. The SSPs are based on five narratives describing alternative socio-economic developments, including sustainable development, regional rivalry, inequality, fossil-fuelled development, and middle-of-the-road development. The long-term demographic and economic projections of the SSPs depict a wide uncertainty range consistent with the scenario literature."	Marker scenario examples		[62]
Drivers	Underlying causes of system change that are external from the system of analysis. They come from higher scales and are not affected by what happens within the system.	Scenario component	Factors	[79], in [149]
Factors	Causes of system change that are internal from the system of analysis. Can be (hopefully) quantified, or at least estimated	Scenario component (internal)		[92]
Factor variables	Discrete elements which are subject to change and have effects on one or more factors	Factor component		[92]
Factor parameters	Discrete elements which are NOT subject to change (possibly based on assumptions and simplifications) and have effects on one or more factors	Factor component		[92]
Trends	An inclination in a particular direction	Attribute of drivers or factors	System development	[92]
Likelihood	The likelihood of an occurrence, an outcome, or a result, where this can be estimated probabilistically	Attribute of drivers or factors	Probability	[149]

13.2. LIST OF RELEVANT POLICY ACTIONS

The following table contains a description of policy actions that are relevant to the future of secondary raw material supply in the EU.

Table 13.2: List of relevant policy actions

POLICY	STATE	YEAR	STATUS	JURISDICTION	LINK
European Institute of Innovation and Technology: Raw Materials Project Call	European Union	2023	Announced	International	
Minerals Security Partnership	European Union	2022	Announced	International	
Resilience for the future: The UK's critical minerals strategy	United Kingdom	2022	In force	National	
Circular Economy Action Plan	Spain	2021	In force	National	
Horizon Europe Strategic Plan (2021 – 2024)	European Union	2021	In force	International	
National Battery Strategy 2025	Finland	2021	In force	National	
National Planning Policy Framework	United Kingdom	2021	In force	National	
EU Sustainable Batteries Regulation	European Union	2020	Announced	Regional	
Green Deal: Circular Economy Action Plan	European Union	2020	In force	International	
Battery fund: 3.2 billion euros for research and innovation	European Union	2019	In force	International	
Resources for France Plan	France	2018	In force	National	
European Battery Alliance	European Union	2017	In force	National	
National Strategy for Energy Research	France	2016	In force	National	
Horizon 2020: Climate action, environment resource efficiency and raw materials	European Union	2013	Ended	International	
Resource Security Action Plan	United Kingdom	2012	In force	National	
Supply of Mineral Resources (SoS MinErals)	United Kingdom	2012	In force	National	
Finland's Minerals Strategy	Finland	2010	In force	National	
Royal Decree 975/2009 about extractive industries waste management and the protection and rehabilitation of areas affected by mining activities	Spain	2009	In force	National	
EU Directive 2006/66/EC Battery Directive	European Union	2006	In force	International	

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13.3. SCENARIO DEVELOPMENT METHODS

Table 13.3 provides an overview of the methods and tools considered, along with a brief description of each and its relevance to the specific context and objectives of the FutuRaM scenario development process.

Table 13.3: Scenario development methods

METHOD	DESCRIPTION	KEY CHARACTERISTICS	LIMITATIONS	APPLICATION
Delphi	Structured expert consultation to gather and distil knowledge and judgments	Iterative rounds of surveys/questionnaires, Expert consensus building	Time-consuming process, May be influenced by dominant opinions or group dynamics	Policy development, Technology foresight, Long-term planning
MCA	Decision-support technique to evaluate and rank scenarios based on criteria	Consideration of multiple dimensions in quantifying qualitative factors	Policy assessment, Project evaluation, Strategic planning	
Forecasting	Use of historical data and statistical methods to predict future trends	Reliance on quantitative models, Time series analysis	Assumption of future patterns based on past data, Sensitivity to data quality and accuracy	Economic forecasting, Demand/supply projections, Financial planning
Backcasting	Working backward from a desired future vision to identify necessary steps	Focus on desired outcomes and future targets, Identification of necessary actions	Uncertainty in future outcomes, Difficulty in determining feasible pathways	Sustainable development planning, Policy design, Long-term goal setting
Scenario Planning	Development of multiple future scenarios to understand the range of possibilities	Identification of key drivers and uncertainties, Narrative construction for each scenario	Subjectivity in scenario construction, Lack of predictive accuracy	Strategic management, Risk assessment, Policy analysis
Morphological Analysis	Exploration of different combinations of variables/factors	Matrix-based exploration of variables and combinations	Complexity in analysing a large number of variables and combinations	Technology assessment, Innovation analysis, System design
Cross-Impact Analysis	Analysis of interdependencies and interactions between variables/factors	Identification of relationships and cross-impacts	Assumptions about causal relationships, Difficulty in capturing complex dynamics	Policy analysis, Risk assessment, System modelling
Morphological Box	Systematic exploration of the potential combinations of different components	Identification of component options and combinations	Complexity in analysing a large number of components and combinations	Technology assessment, Innovation analysis, Decision-making
Gausemeier approach	Scenario development method involving the identification of future developments, evaluation of influencing factors, and determination of desired and undesired developments	Systematic analysis of future developments and factors	Relies on expert judgment and subjective assessments	Strategic planning, Innovation management

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METHOD	DESCRIPTION	KEY CHARACTERISTICS	LIMITATIONS	APPLICATION
Schwartz' 8-Step Scenario Model	Scenario building model consisting of eight steps: identify the focal issue, determine the key forces, construct the scenario framework, identify driving forces, assess the uncertainties, develop the scenarios, analyze the scenarios, and monitor and adjust the scenarios	Systematic progression through stages of scenario development	Requires detailed data and analysis	Strategic planning, Decision-making
Schoemaker's 10-Step Scenario Model	Scenario building model consisting of ten steps: identify the focal issue, determine the scope, identify the key driving forces, develop the scenarios, define the scenario logic, assess the scenarios, refine the scenarios, examine implications, formulate actions, and communicate results	Emphasis on thorough analysis and evaluation of scenarios	Can be time-consuming and resource-intensive	Strategic planning, Risk management

13.4. MARKER SCENARIO MAPPING

Table 13.4 below presents an overview of the marker scenarios that were considered in the scenario development phase of the FutuRaM project. The table is not intended to be exhaustive, but rather to provide an overview of the different scenarios that have been developed in the field of waste management, resource recovery, and circular economy.

Table 13.4: Overview of marker scenarios

LITERATURE	TYPE	WASTE STREAM	TEMPORAL COVERAGE	LOCATION	NUMBER OF SCENARIOS	LINK
The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview	Academic	All (narratives)	Scenario to 2100	Global	5 SSPs	
Environmental Impacts of Global Offshore Wind Energy Development until 2040	Academic	CDW	Scenario: 2019–2040	Global	4 (based on IEA)	
Global greenhouse gas emissions from residential and commercial building materials and mitigation strategies to 2060	Academic	CDW	Scenario: 2020–2060	Global	2 (based on SSP2)	
Modelling global material stocks and flows for residential and service sector buildings towards 2050	Academic	CDW	Scenario: 2020–2060	Global	1 (SSP2)	
The evolution and future perspectives of energy intensity in the global building sector 1971–2060	Academic	CDW	Scenario: 2020–2060	Global	1 (SSP2)	
Tracking Construction Material over Space and Time Prospective and Geo-referenced modelling of Building Stocks and Construction Material Flows	Academic	CDW	Scenario to 2060	Global	6 scenarios concerning per-capita floor area, building stock turnover, and construction material.	
Global construction materials database and stock analysis of residential buildings between 1970–2050	Academic	CDW	Scenario to 2060	Global	1 (SSP2)	
A comprehensive set of global scenarios of housing, mobility, and material efficiency for material cycles and energy systems modelling	Academic	CDW	Scenario to 2060	Global	Low energy demand, SSP1, SSP2	
Global scenarios of resource and emission savings from material efficiency in residential buildings and cars	Academic	CDW, ELV	Scenarios to 2050	Global	SSP1, SSP2	
Matching global cobalt demand under different scenarios for co-production and mining attractiveness	Academic	BAT	2050	Global	5	
Copper at the crossroads: Assessment of the interactions between low-carbon energy transition and supply limitations	Academic	Copper	2050	Global	2: 2°C and 4°C	
The impact of climate policy implementation on lithium, cobalt and nickel demand: The case of the Dutch automotive sector up to 2040	Academic	ELV, Batteries	Scenario: 2019–2040	NL	2 (Based on policies)	
The rise of electric vehicles—2020 status and future expectations	Academic	ELV, BAT	up to 2050	Global	various	

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LITERATURE	TYPE	WASTE STREAM	TEMPORAL COVERAGE	LOCATION	NUMBER OF SCENARIOS	LINK
Scenarios for the Return of Lithium-ion Batteries out of Electric Cars for Recycling	Academic	ELV, Battery	Scenario to 2050	Global	2	🔗
The dynamic equilibrium mechanism of regional lithium flow for transportation electrification	Academic	ELV, BAT	Scenario to 2050	Global	1 (projection)	🔗
Future material demand for automotive lithium-based batteries	Academic	ELV, BAT	Scenario to 2050	Global	4 (based on IEA)	🔗
Analysis of the Li-ion battery industry in light of the global transition to electric passenger light-duty vehicles until 2050	Academic	ELV, BAT	Scenario to 2050	Global	Combination of SSPs and RCPs	🔗
Circular economy strategies for electric vehicle batteries reduce reliance on raw materials	Academic	ELV, BAT	Scenario to 2050	Global	Reference + 4 technologies	🔗
Summary and critical review of the International Energy Agency's special report: The role of critical minerals in clean energy transitions	Academic	Energy	2050	Global	n/a	🔗
Review of critical metal dynamics to 2050 for 48 elements	Academic	Energy	Scenario to 2050	Global	1 compiled from various renewable technologies	🔗
Major metals demand, supply, and environmental impacts to 2100: A critical review	Academic	Energy	Scenario to 2100	Global	1 review of 197 studies	🔗
Requirements for Minerals and Metals for 100% Renewable Scenarios	Academic	Energy	Scenario to 2050	Global	1.5 degree scenario	🔗
The 3-machines energy transition model: Exploring the energy frontiers for restoring a habitable climate	Academic	Energy	2100	Global	20, rapid transition stabler 1.5 °C and return to 350 ppm	🔗
Modelling the demand and access of mineral resources in a changing world	Academic	Energy, Construction	2060	Global	RTS, BD2S IEA	🔗
Rare earths in the energy transition: what threats are there for the 'vitamins of modern society'?	Academic	Rare earths	2050	Global	2: 2°C and 4°C	🔗
A slag prediction model in an electric arc furnace process for special steel production	Academic	SLASH	None	Global	n/a	🔗
Decarbonising the iron and steel sector for a 2°C target using inherent waste streams	Academic	SLASH	Scenario to 2050	Global	1 (2 degree climate goal)	🔗
Environmental Implications of Future Demand Scenarios for Metals: Methodology and Application to the Case of Seven Major Metals	Academic	Various	Scenario to 2050	Global	4 (UN GEO-4)	🔗

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LITERATURE	TYPE	WASTE STREAM	TEMPORAL COVERAGE	LOCATION	NUMBER OF SCENARIOS	LINK
Resource Demand Scenarios for the Major Metals	Academic	Various	Scenario to 2050	Global	4 (UN GEO-4)	
Raw material depletion and scenario assessment in European Union — A circular economy approach	Academic	Various	None	EU	n/a	
Material bottlenecks in the future development of green technologies	Academic	Various	Scenario to 2050	Global	1 (BAU)	
Reuse assessment of WEEE: Systematic review of emerging themes and research directions	Academic	WEEE	None	Global	n/a	
A systematic literature review on the circular economy initiatives in the European Union	Academic	Circularity	None	EU	Circular strategies	
Material Flow Accounting: Measuring Global Material Use for Sustainable Development	Academic	Various	Scenario to 2100	Global	1 (BAU)	
Circular Economy Action Plan	Action plan	Various	Scenario to 2050	EU	35 actions to climate neutrality	
Construction and demolition waste: challenges and opportunities in a circular economy	Report	CDW	None	EU	n/a	
IEA world energy model	Report	Energy	Scenario to 2050	Global	4	
Bloomberg scenarios	Report	Energy	Scenario to 2050	Global	3	
The Role of Critical Minerals in Clean Energy Transitions	Report	Energy	None	Global	n/a	
Transitions to 2050 decide now act for climate	Report	Energy	Scenario to 2050	France	4 to reach 2.1C by 2100	
Raw materials demand for wind and solar PV technologies in the transition towards a decarbonised energy system	Report	Energy	Scenario to 2050	EU	low and high material demand scenarios	
Inventaires des besoins en matière, énergie, eau et sols des technologies de la transition énergétique	Report	Energy	Scenario to 2050	France	1	
Minerals in the future of Europe	Report	MinW	Scenario to 2050	EU	3 (2050 net-zero, digital, circular)	
Minerals, Critical Minerals and the US Economy	Report	MinW	None	US	n/a	
Minéraux stratégiques — État des lieux et propositions pour une vision partagée	Report	MinW	None	FR	n/a	

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LITERATURE	TYPE	WASTE STREAM	TEMPORAL COVERAGE	LOCATION	NUMBER OF SCENARIOS	LINK
The Critical Raw Materials (CRM) initiative — Underpinning the strategic approach to the EU's raw materials policy	Report	MinW	None	EU	n/a	
Towards the Circular Economy: Accelerating the scale-up across global supply chains	Report	Circularity	None	Global	n/a	
The Circular Economy in Europe	Report	Circularity	None	EU	n/a	
Global material flows and resource productivity: Forty years of evidence	Report	Circularity	None	Global	n/a	
The circular economy concept: contextualisation and multiple perspectives	Report	Circularity	None	Global	n/a	
Global material flows database	Database	Various	None	Global	n/a	
International Resource Panel	Reports	Various	None	Global	n/a	
World Business Council for Sustainable Development	Reports	Various	None	Global	n/a	
Ellen MacArthur Foundation	Reports	Various	None	Global	n/a	
European Environment Agency	Reports	Various	None	EU	n/a	
International Energy Agency	Reports	Energy	None	Global	n/a	
United Nations Environment Programme	Reports	Various	None	Global	n/a	
United Nations Industrial Development Reports	Reports	Various	None	Global	n/a	
World Bank	Reports	Various	None	Global	n/a	
World Economic Forum	Reports	Various	None	Global	n/a	

13.5. DRIVERS AND FACTORS IDENTIFIED IN THE INITIAL COLLECTION PHASE

Table 13.5 lists the elements that were identified in the initial phase of driver/factor collection.

Table 13.5: Drivers and factors identified in the initial collection phase

METHOD	DESCRIPTION
Stricter environmental regulations	Increased regulations and policies aimed at reducing environmental impact
Inflation	Increase in the general price level of goods and services over time
Employment rates	Percentage of the working-age population that is employed
Exchange rates	Value of one currency relative to another currency
Interest rates	Cost of borrowing money or the return on investment
Gasoline price	Cost of gasoline for vehicles
Electricity price	Cost of electricity for consumers or businesses
Raw material prices	Prices of primary materials used in production processes
CO2 market	Trading system for carbon emissions permits or credits
Education level	Level of education attained by individuals or the overall population
Volunteering	Engagement in unpaid activities for the benefit of others
Transparency	Openness, accountability, and information accessibility
Compliance with rules	Adherence to regulations, guidelines, or standards
Cultural values / Consciousness	Beliefs, attitudes, and awareness of individuals and society
Accessibility	Ease of access to goods, services, or infrastructure
Land rights	Legal rights to ownership, use, or access to land
Work-life balance	Equilibrium between work and personal life
Urbanisation	Increase in the population living in urban areas
Water supply constraints	Limitations on the availability or access to freshwater resources
Increased intrinsic drive for env. protection	Growing internal motivation to protect and conserve the environment
NIMBY to projects	Not-In-My-Backyard opposition to the location of certain projects
Climate change impacts (flooding, etc.)	Consequences of climate change, such as increased flooding or extreme events
Climate change mitigation efforts	Actions taken to reduce greenhouse gas emissions and combat climate change
Redundancy	Availability of backup systems or alternative options
Material efficiency	Effective use and management of materials to minimize waste and loss
Energy efficiency of buildings	Performance and efficiency of energy consumption in buildings

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METHOD	DESCRIPTION
Change of products in the scope WEEE directive	Inclusion or exclusion of certain products within the scope of the WEEE directive
GDP/PPP	Gross Domestic Product (GDP) adjusted for purchasing power parity (PPP)
Improved repairability	Enhanced ability to repair and maintain products or equipment
Target enforcement	Implementation and enforcement of specific targets or goals
Data protection	Safeguarding personal data and ensuring privacy
Infrastructure	Physical structures and facilities necessary for the functioning of society
Intellectual property issues	Legal rights and protections for intellectual creations and innovations
Population	Total number of people in a given area or region
Resource shortage	Insufficient availability or scarcity of natural resources
Treatment cost	Cost of waste treatment, disposal, or recycling processes
Digital product passports	Digital documentation providing information about a product's lifecycle
Obsolescence	State of being outdated or no longer in use or demand
Digitalization	Integration and adoption of digital technologies and processes
SRM prices	Prices of secondary raw materials or recycled materials
Product prices	Prices of goods or products in the market
Recyclability mandates	Requirements or regulations promoting the recyclability of products
Conflict in supply chain	Disputes or conflicts within the supply chain of raw materials or products
Obligatory recycling standards for treatment facilities	Mandatory standards for recycling processes in treatment facilities
Improved durability	Enhanced longevity and resistance of products or materials
Composition change	Alteration or modification of the composition of materials or products
Subsidies	Financial support or incentives provided by governments or organizations
Availability of recovery technologies	Existence and accessibility of technologies for material recovery
Taxation (raw materials, landfill)	Imposition of taxes on raw materials or landfill activities
Obligatory removal of CRMs from waste	Required removal or extraction of critical raw materials from waste streams
Corruption	Dishonest or unethical behaviour, typically involving misuse of power
Supply chain due diligence laws	Regulations or laws requiring companies to assess and manage supply chain risks

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METHOD	DESCRIPTION
Improved recyclability	Increased ability of products or materials to be recycled or reused
Ecodesign	Designing products with consideration for their environmental impact
Trade barriers	Barriers or restrictions to international trade or commerce
Industrialization of Europe	Development and growth of industrial activities in European countries
Reduced consumerism	Shift towards decreased consumption and a more sustainable lifestyle
Accessibility/Infrastructure	Availability and adequacy of infrastructure to support accessibility
New mines in rich EU countries?	Establishment of new mines in economically prosperous European countries
Miniaturization	Process of making products or components smaller and more compact
Sharing economy	Economic system based on sharing resources and services
Repairability mandates	Requirements or regulations promoting the repairability of products
Renewable energy targets	Set goals or objectives for increasing the use of renewable energy sources

13.6. DRIVERS AND FACTORS IDENTIFIED IN THE SCREENING PHASE

The following table lists the scenario elements that were identified in the screening phase of driver/factor collection.

Table 13.6: List of drivers and factors identified in the screening phase

DOMAIN	DRIVER/FACTOR	DEFINITION	BAU	REC	CIR
Economic	CO2 market price	Price of carbon dioxide (CO2) emissions in carbon markets	I	I	I
Economic	Economic growth	Rate of economic growth	I	I	I
Economic	Energy prices	Prices of energy resources	I	I	I
Economic	Market saturation	Level of saturation reached in the market for certain products or services	I	I	II
Economic	Raw material vs SRM prices	Price comparison between raw materials and Secondary Raw Materials (SRMs)	I	I	I
Economic	Re-industrialisation of EU	Process of revitalizing industrial activities in the European Union	I	I	I
Environmental	Climate change impacts (flooding, etc.)	Impacts of climate change such as flooding and other related events	I	I	I
Environmental	Climate change mitigation efforts	Efforts made to mitigate the effects of climate change	I	I	I
Environmental	Increased drive for env. protection	Growing motivation and drive to protect the environment	I	III	III
Environmental	Resource shortage	Shortage of natural resources	I	I	I
Legal/Political	Ecodesign/re-X mandates	Establishment of ecodesign requirements for specific product groups to improve circularity, energy performance, and other environmental sustainability aspects	I	II	III
Legal/Political	Governance: corruption vs compliance	Contrasting levels of corruption and compliance within governance systems	I	I	I
Legal/Political	International trade and co-operation (vs. autarky)	Level of international trade and cooperation versus self-sufficiency	I	I	I
Legal/Political	Product information transparency	Provision of transparent product information to consumers, manufacturers, importers, repairers, recyclers, or national authorities	I	III	III
Legal/Political	Progress toward renewable energy targets	Progress made in achieving renewable energy targets	I	I	I
Legal/Political	Stricter environmental regulations	Implementation of more stringent rules and regulations for environmental protection	I	III	III
Legal/Political	Subsidies/taxation to promote circularity	Provision of subsidies or implementation of taxation policies to incentivize circularity	I	I	I
Legal/Political	Supply chain due diligence laws	Implementation and enforcement of laws requiring companies to address negative human rights and environmental impacts in their value chains	I	II	III
Social	Hoarding	The act of stockpiling and keeping excessive amounts of products	III	II	II
Social	NIMBY to projects	Opposition of local communities to the location of new projects, such as mining, in their vicinity	I	I	I
Social	Participation in re-X activities	"Involvement in activities related to the ""re-"" concepts, including refusing, reducing, repairing, and reusing products"	I	II	III
Social	Population	Size and growth of the population	I	I	I

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DOMAIN	DRIVER/FACTOR	DEFINITION	BAU	REC	CIR
Social	Urbanisation	Rate of urban population growth	I	I	I
Technical	Digitisation	Adoption and integration of digital technologies	I	I	I
Technical	Integration of SRM system across EU	Integration of a Secondary Raw Materials (SRM) system across the European Union	I	III	III
Technical	Product technology	Changes in product function or composition that lead to changes in waste stream composition and quantity	I	III	III
Technical	Recovery technology	Technologies and processes for recovering materials from waste	I	III	III

13.7. DRIVERS AND FACTORS AFTER CATEGORISATION

The 21 elements that were identified in this stage are listed in Table 13.7.

Note that CIR and REC are very similar for many scenario elements, the main difference being the way in which the targets are achieved. That is, for CIR, re-X strategies are promoted, whereas, for REC, the focus is on technological advancements in the recovery system.

This distinction will have a significant impact on how the scenarios are quantitatively modelled and on the subsequent outcomes of these models.

Table 13.7: List of drivers and factors identified in the screening phase

DOMAIN	DRIVER/FACTOR	DEFINITION	INTERNAL	BAU	REC	CIR
TECH	Recovery technology	Implementation and advancements in waste recovery technologies	TRUE	I	III	III
TECH	Product technology	Changes in product function or composition	TRUE	I	III	III
TECH	Integration of SRM system across EU	Integration of a secondary raw material recovery system across EU countries	TRUE	I	III	III
ENV	Increased drive for environmental protection	Growing concern and motivation for environmental conservation	TRUE	I	III	III
ECO	Progress toward renewable energy targets	Advancements and achievements in renewable energy generation	TRUE	III	III	III
ECO	Subsidies and taxation to promote circularity	Financial incentives or taxes to encourage circular economy	TRUE	I	II	III
SOC	Participation in re-X activities	Engagement in refuse-reduce-repair-reuse activities	TRUE	I	I	III
POL	Stricter environmental regulations	Tightening of environmental laws and regulations	TRUE	II	III	III
POL	Stricter waste management regulations	Strengthening of waste management laws and regulations	TRUE	II	III	III
POL	Supply chain due diligence laws: implementation and enforcement	Obligations for identifying and mitigating negative impacts in supply chains	TRUE	I	III	III
POL	Compliance with waste targets	Meeting specific waste management and recycling targets	TRUE	I	III	III
ENV	Resource shortages	Limited availability of natural resources	FALSE	n/a	n/a	n/a
ECO	Raw material vs SRM prices	Price dynamics and competition between raw materials and secondary raw materials	FALSE	n/a	n/a	n/a
ENV	Climate change impactsmitigation	Effects and actions related to climate change	FALSE	n/a	n/a	n/a
ECO	International trade and co-operation (vs. autarky)	Collaborative trade agreements and global cooperation	FALSE	n/a	n/a	n/a
ECO	Energy prices	Costs and fluctuations in energy prices	FALSE	n/a	n/a	n/a
ECO	Economic growth	Overall economic expansion and development	FALSE	n/a	n/a	n/a
ECO	Re-industrialisation of EU	Shift towards increased industrial activities in the EU	FALSE	n/a	n/a	n/a
SOC	NIMBY to projects	Opposition to local projects and developments	FALSE	n/a	n/a	n/a

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DOMAIN	DRIVER/FACTOR	DEFINITION	INTERNAL	BAU	REC	CIR
SOC	Population and urbanisation	Growth and urban development of population	FALSE	n/a	n/a	n/a
ECO	CO2 market price	Price and market dynamics of carbon emissions	FALSE	n/a	n/a	n/a

13.8. DRIVERS AND FACTORS FOR QUANTIFICATION

The following Table 13.8 lists the categorised scenario elements that were quantified and incorporated into the modelling.

Table 13.8: List of scenario elements categorised for quantification. The roman numerals in the columns BAU, REC, and CIR represent the magnitude of the future trend for the element in the scenario. Internal, external, and outside refer to the classification type of the scenario element

DOMAIN	ELEMENT	INTERNAL	EXTERNAL	OUTSIDE	BAU	REC	CIR	PARAMETERS AFFECTED
ECO	Subsidies and taxation to promote circular strategies	✓			I	I	III	demand, waste generation, lifetimes, sharing, collection,
POL	Targets and enforcement to promote circular strategies	✓			I	I	III	demand, waste generation, lifetimes, sharing, collection
SOC	Participation in re-X activities	✓			I	I	III	demand, waste generation, lifetimes, sharing, collection,
ECO	Subsidies and taxation to promote recovery strategies	✓			I	III	I	recycling rates, recovery capacity, recovery impacts, collection
POL	Targets and enforcement to promote recovery strategies	✓			I	III	I	recycling rates, recovery rates, capacity
POL	Supply chain due diligence laws	✓			I	III	III	composition, export
POL	Stricter environmental regulations	✓			I	III	III	composition, waste generation, lifetimes, export,recovery rates, recovery capacity, recovery impacts
POL	Stricter waste management regulations	✓			I	III	III	composition, waste generation, lifetimes, export, recovery rates, recovery capacity, recovery impacts
TECH	Product technology	✓			I	III	III	lifetimes, recovery rates, recovery impacts
TECH	Recovery technology	✓			I	III	III	recovery rates, recovery capacity, recovery impacts
TECH	Integration of SRM recovery system across Europe	✓			I	III	III	recycling rates, recovery rates, recovery capacity, recovery impacts
ECO	Progress toward renewable energy targets		✓		-	-	-	composition, demand, waste generation, recovery impacts
ECO	Economic growth		✓		-	-	-	composition, demand, waste generation
SOC	Population		✓		-	-	-	demand, waste generation
ECO	Primary vs. secondary raw material prices		✓		n/a	n/a	n/a	considered in sensitivity analysis

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DOMAIN	ELEMENT	INTERNAL	EXTERNAL	OUTSIDE	BAU	REC	CIR	PARAMETERS AFFECTED
ECO	Energy prices		✓		n/a	n/a	n/a	considered in sensitivity analysis
ECO	Carbon price		✓		n/a	n/a	n/a	considered in sensitivity analysis
ENV	Resource supply constraints		✓		n/a	n/a	n/a	considered in sensitivity analysis:
ECO	International trade and co-operation (vs. autarky)			✓	n/a	n/a	n/a	not model input (resource supply constraints is a proxy)
ECO	Re-industrialisation of EU			✓	n/a	n/a	n/a	not model input
SOC	Resistance to recovery projects (NIMBY)			✓	n/a	n/a	n/a	not model input (considered in UNFC assessments)